

BFS Traversal

Step 1: Start at A

Legend

Complete

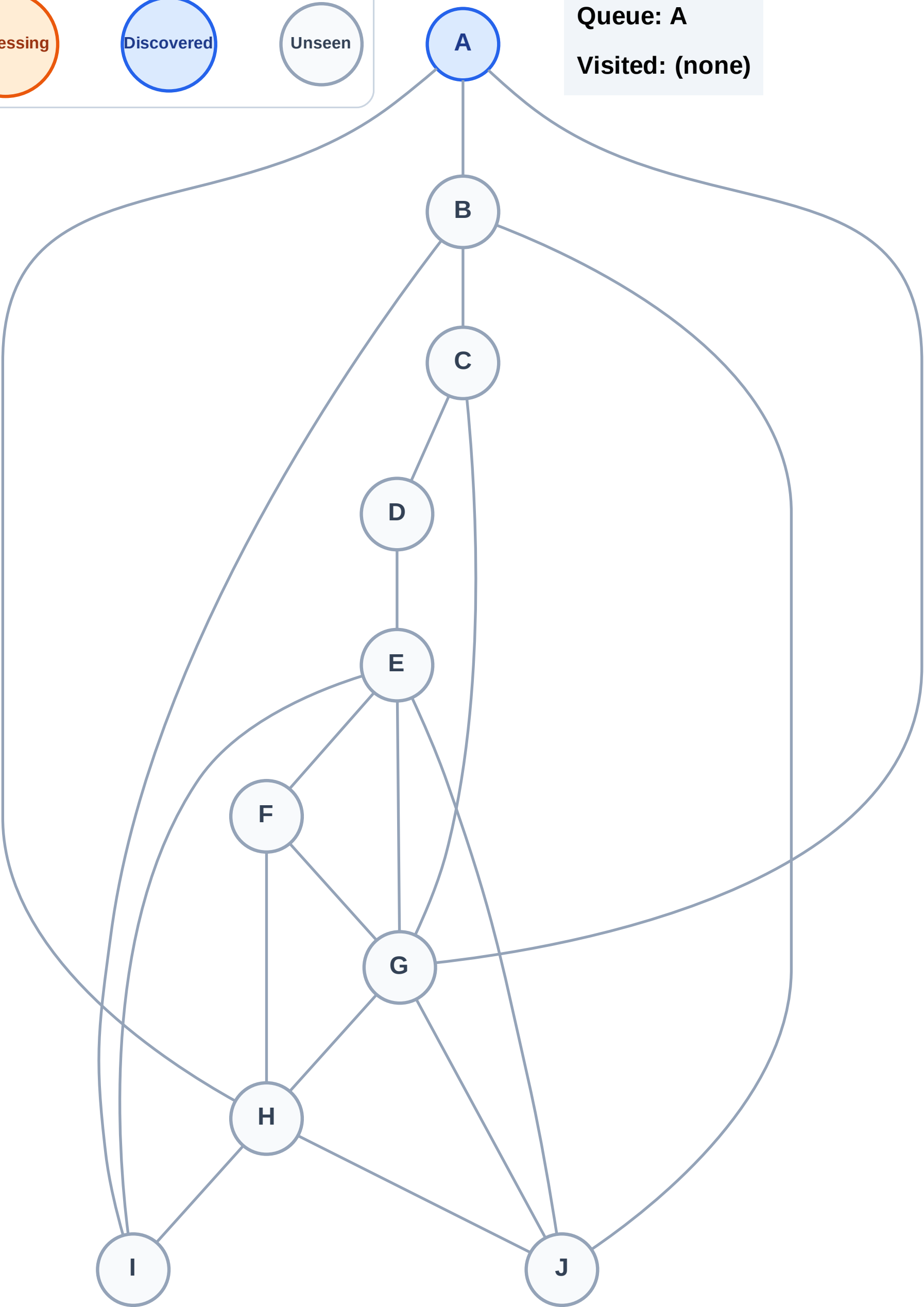
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

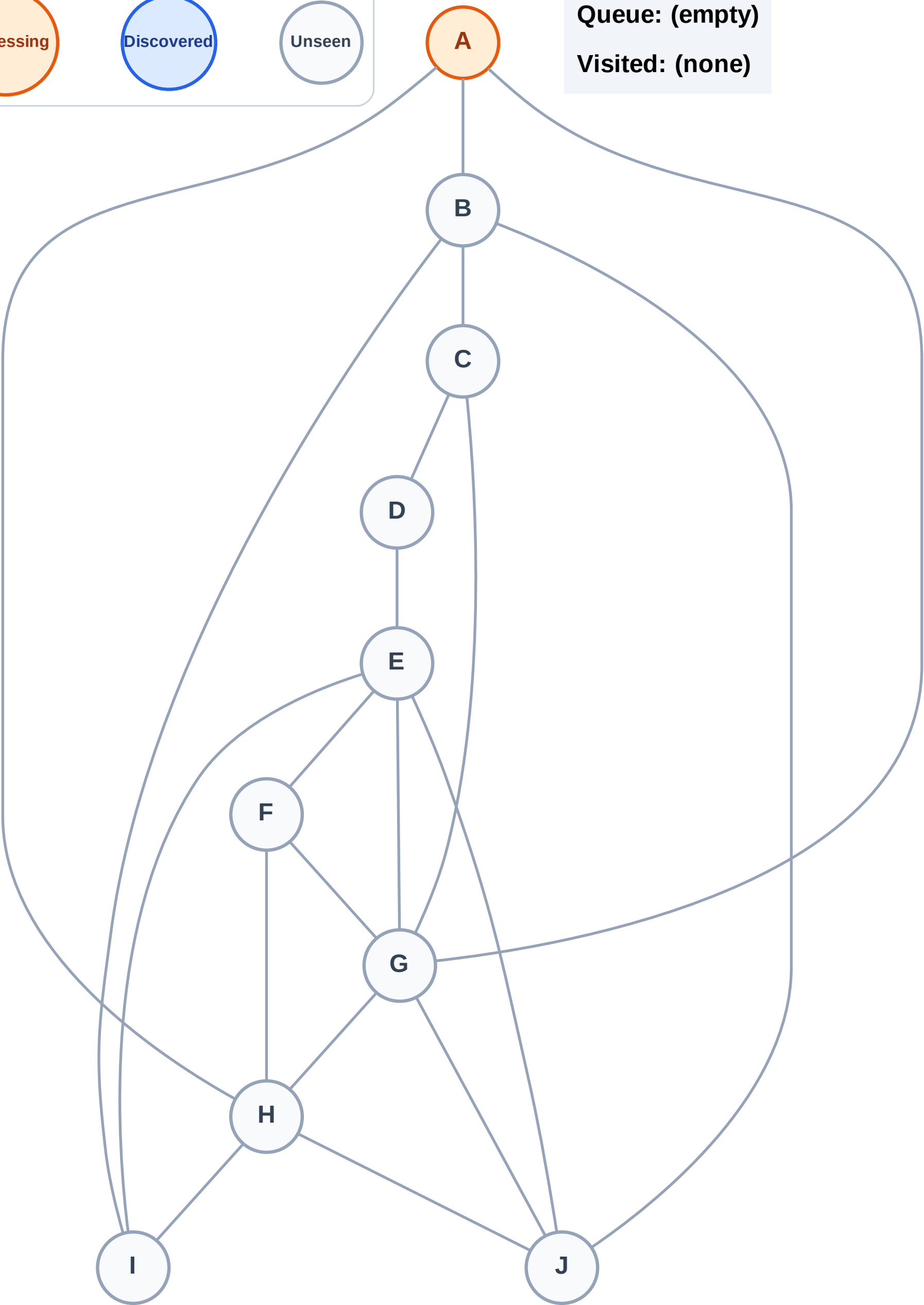
Complete

Processing

Discovered

Unseen

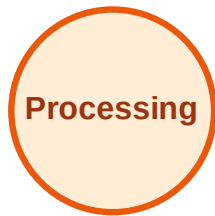
Queue: (empty)
Visited: (none)



BFS Traversal

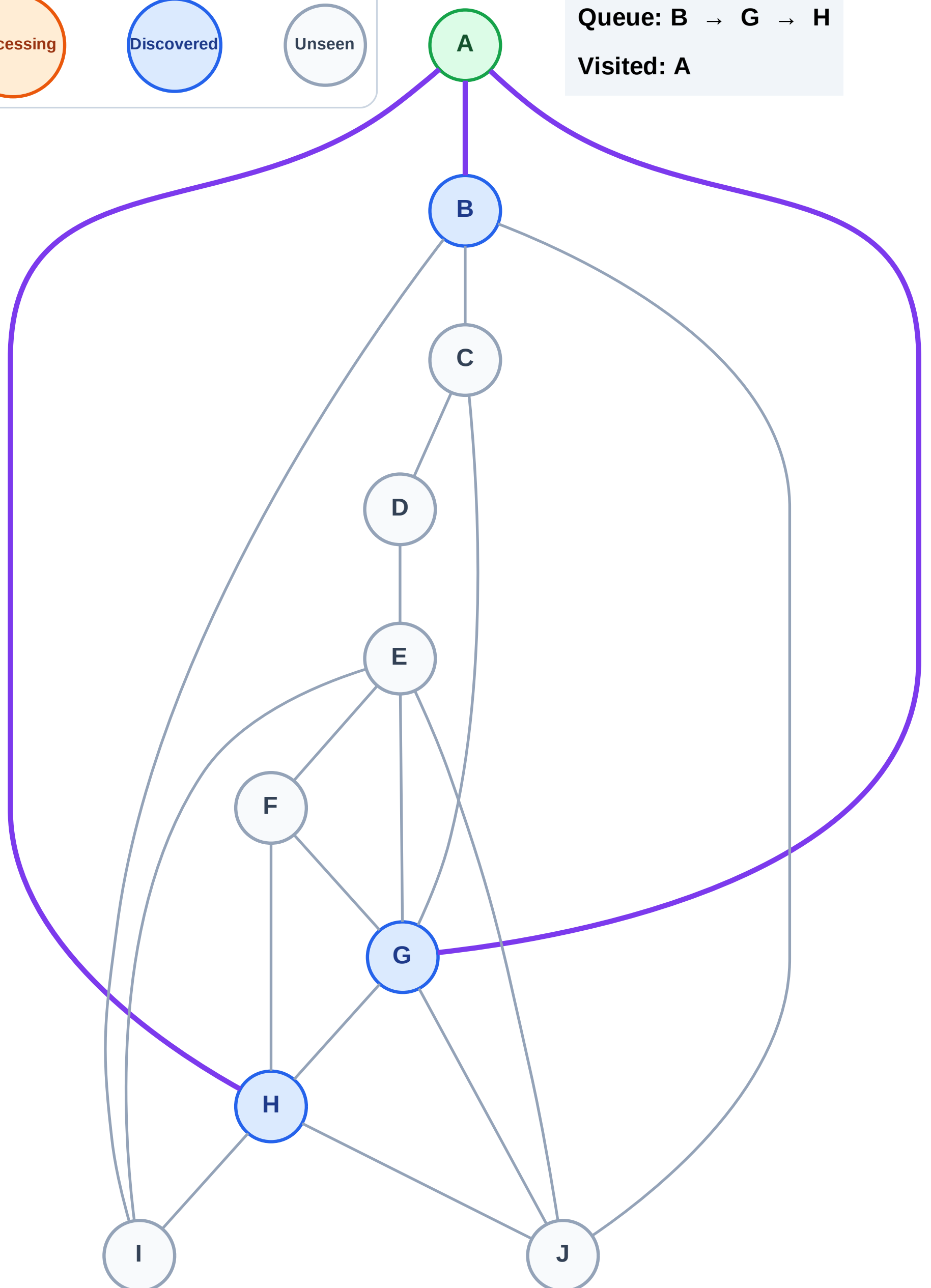
Step 3: Discover B, G, H from A

Legend



Queue: B → G → H

Visited: A



BFS Traversal

Step 4: Process node B

Legend

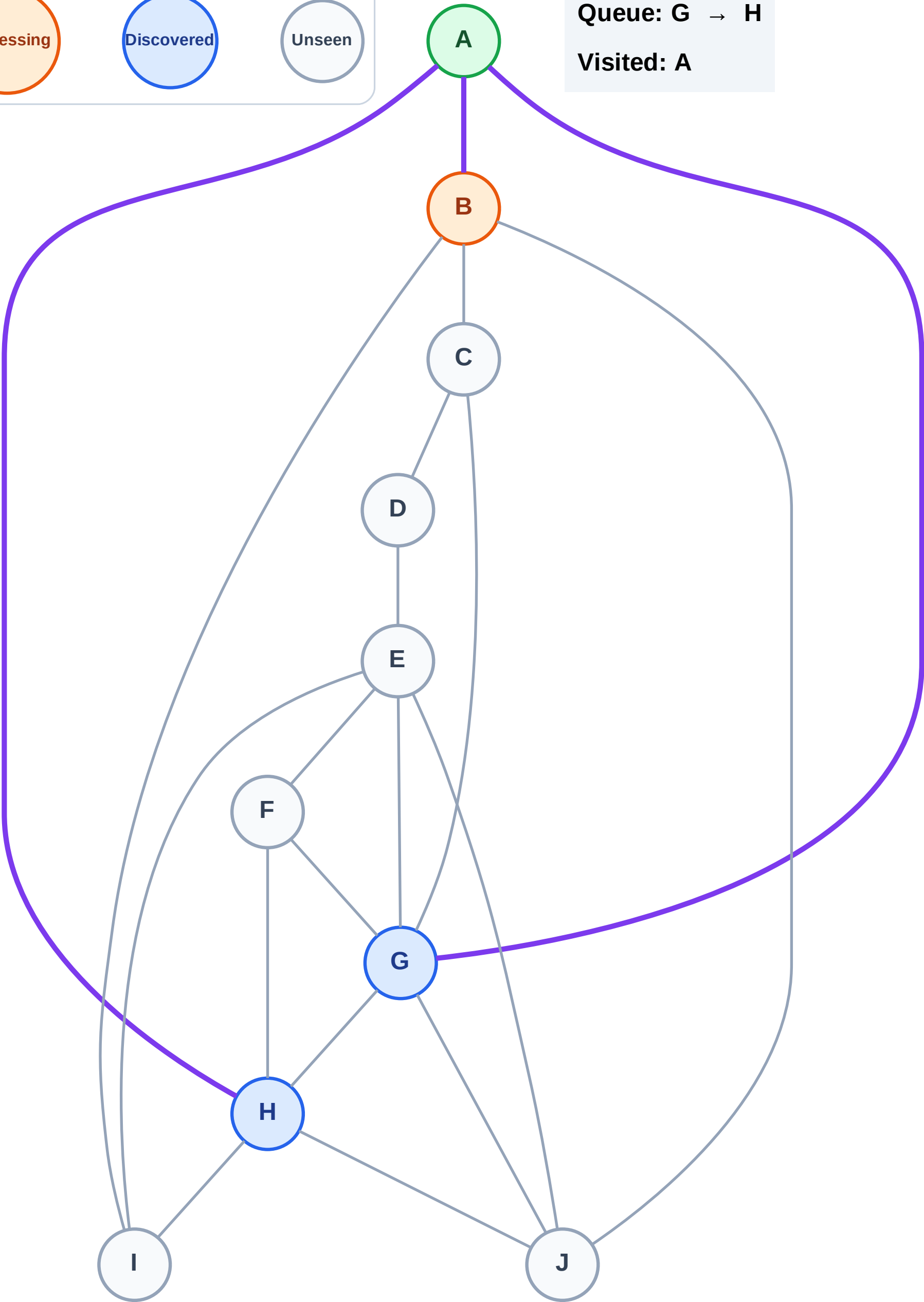
Complete

Processing

Discovered

Unseen

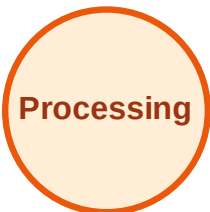
Queue: G → H
Visited: A



BFS Traversal

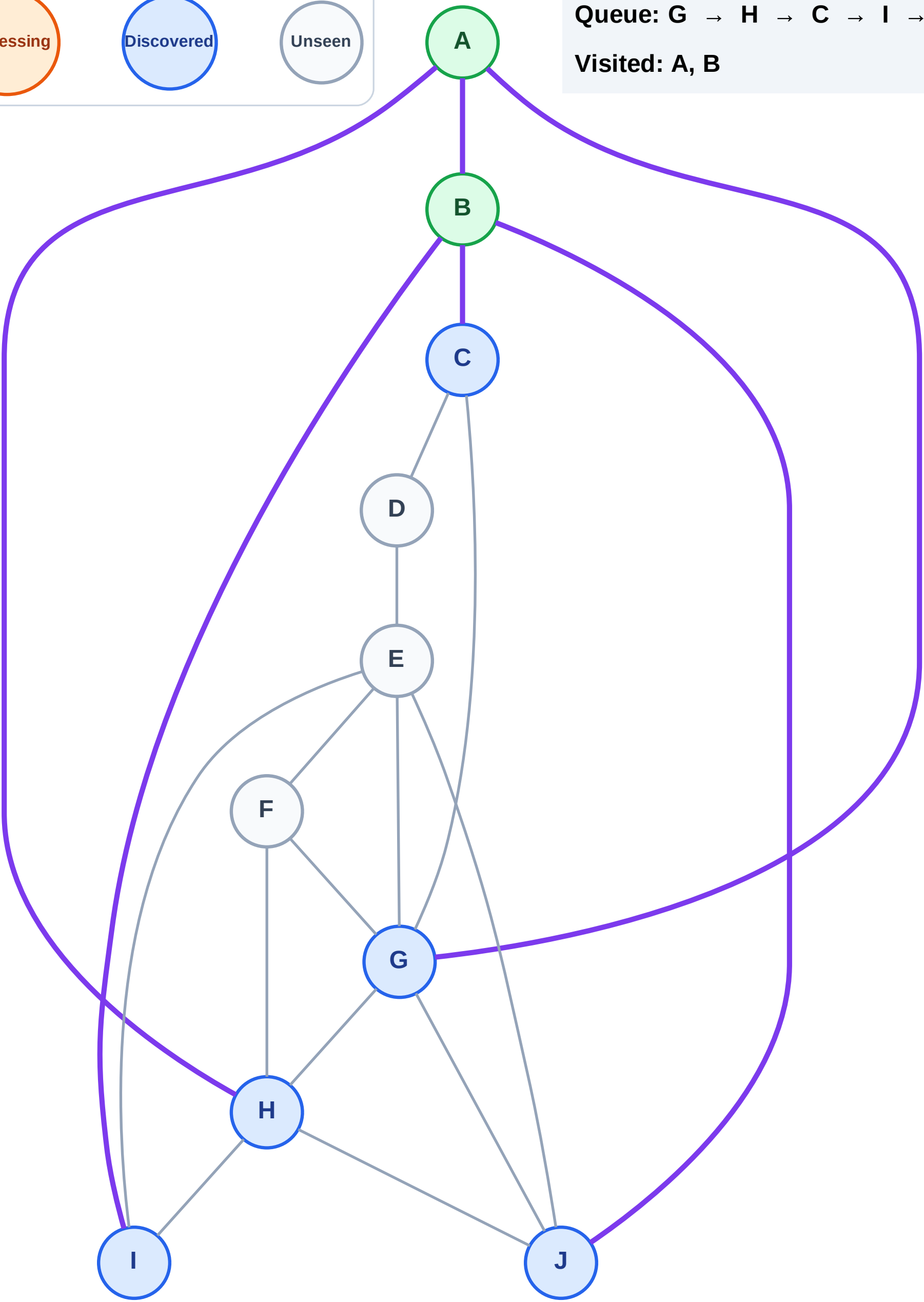
Step 5: Discover C, I, J from B

Legend



Queue: G → H → C → I → J

Visited: A, B



BFS Traversal

Step 6: Process node G

Legend

Complete

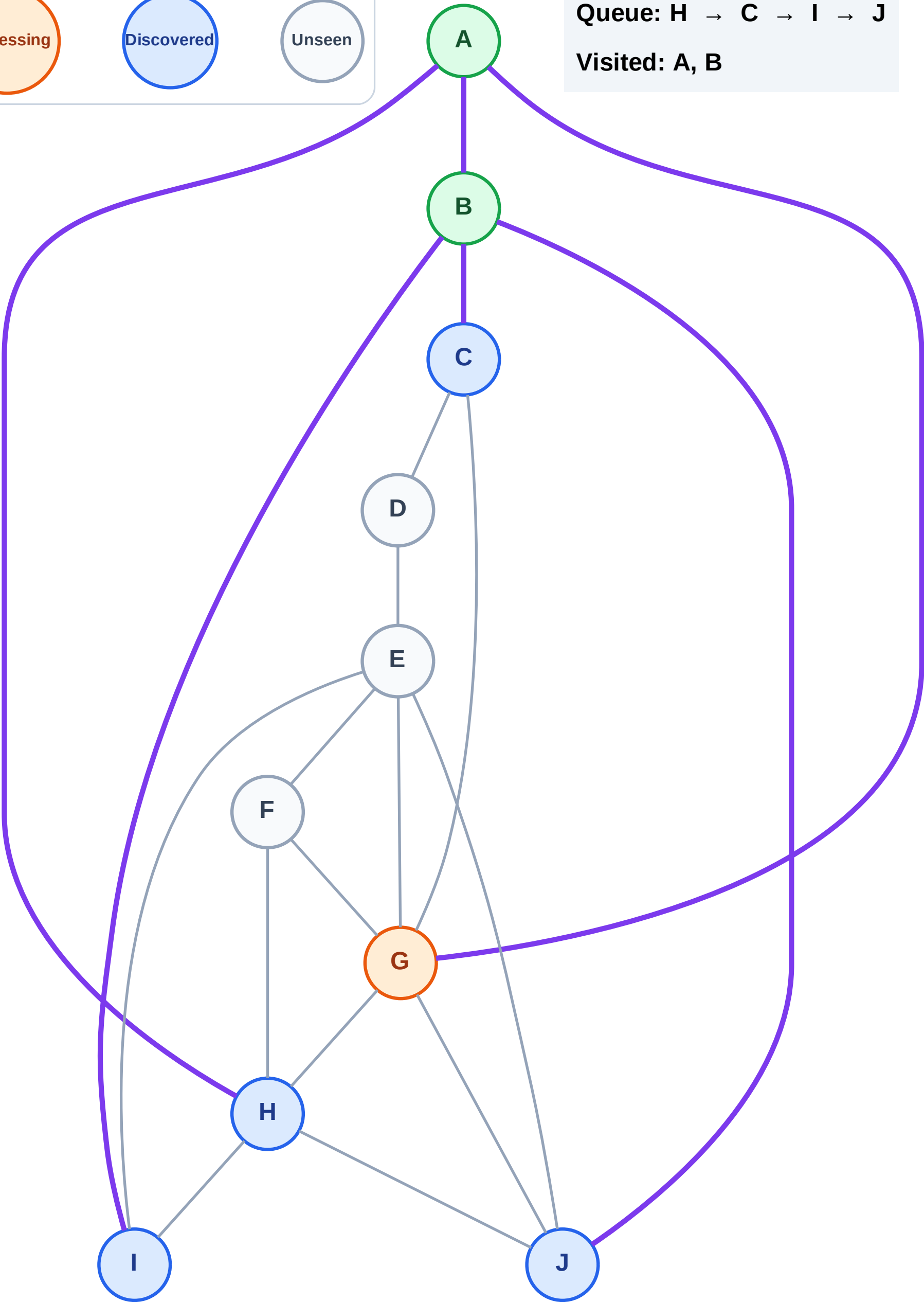
Processing

Discovered

Unseen

Queue: H → C → I → J

Visited: A, B



BFS Traversal

Step 7: Discover E, F from G

Legend

Complete

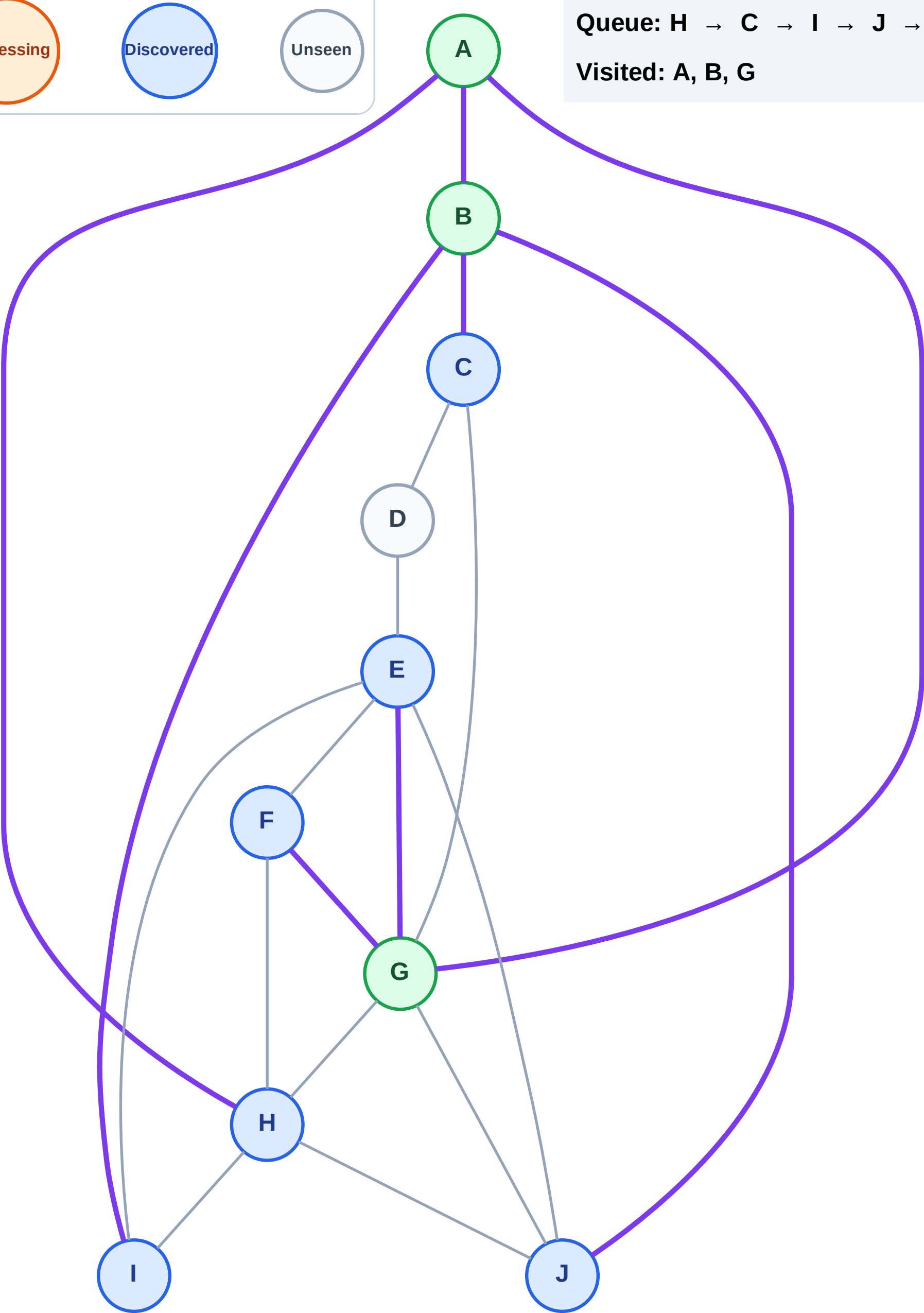
Processing

Discovered

Unseen

Queue: H → C → I → J → E → F

Visited: A, B, G

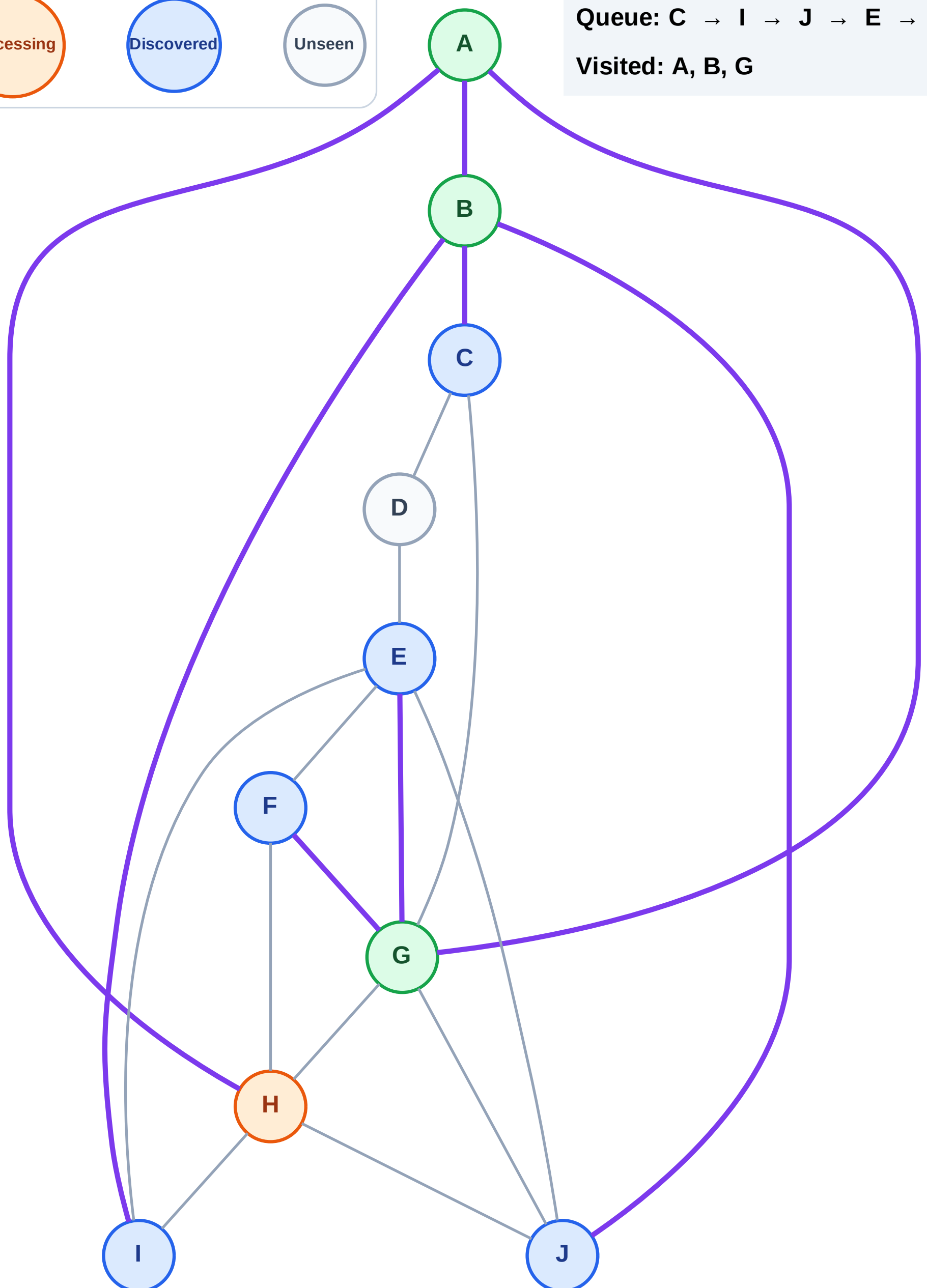


Step 8: Process node H

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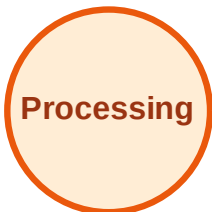
Visited: A B C



BFS Traversal

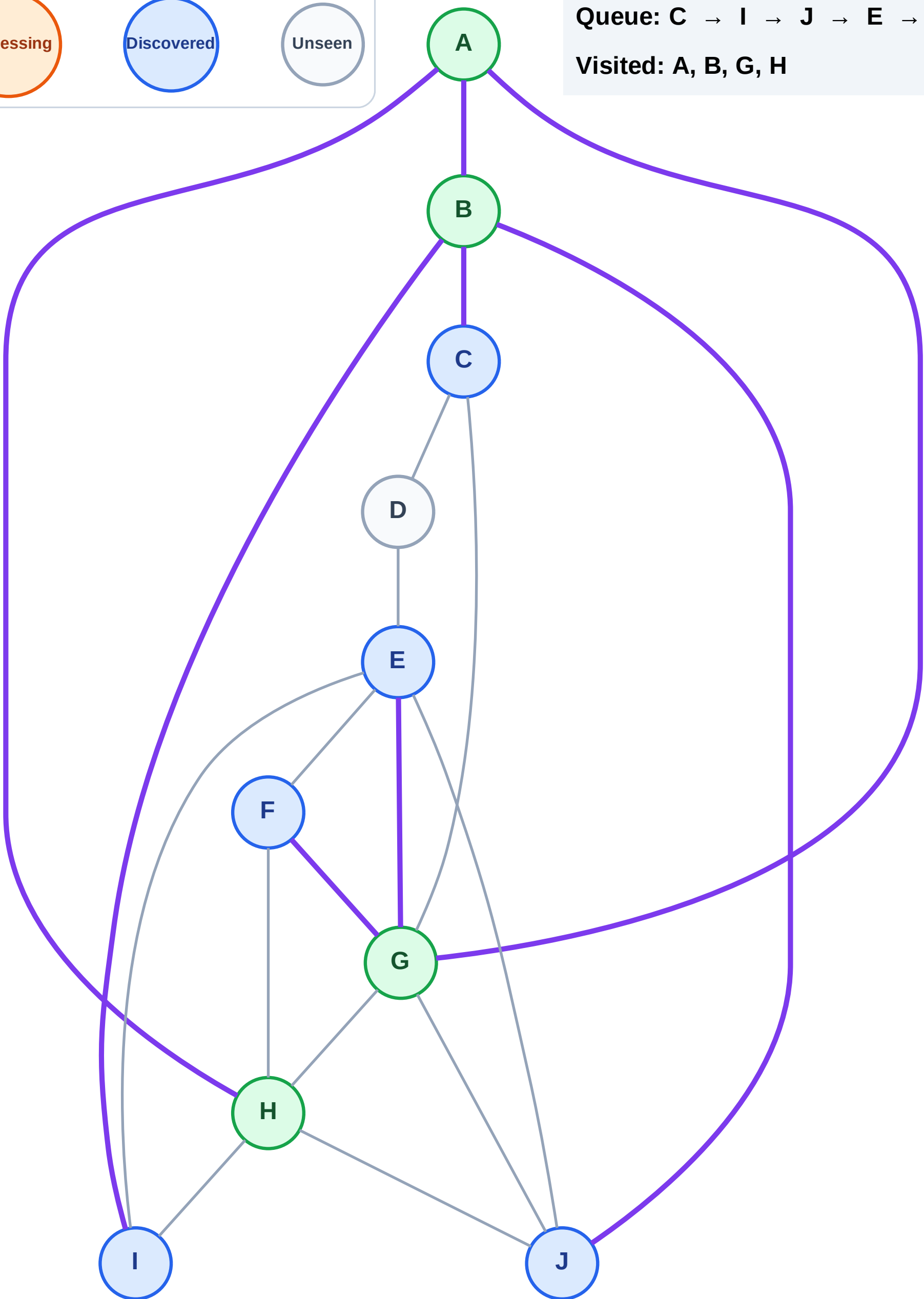
Step 9: No new neighbors from H

Legend



Queue: C → I → J → E → F

Visited: A, B, G, H



BFS Traversal

Step 10: Process node C

Legend

Complete

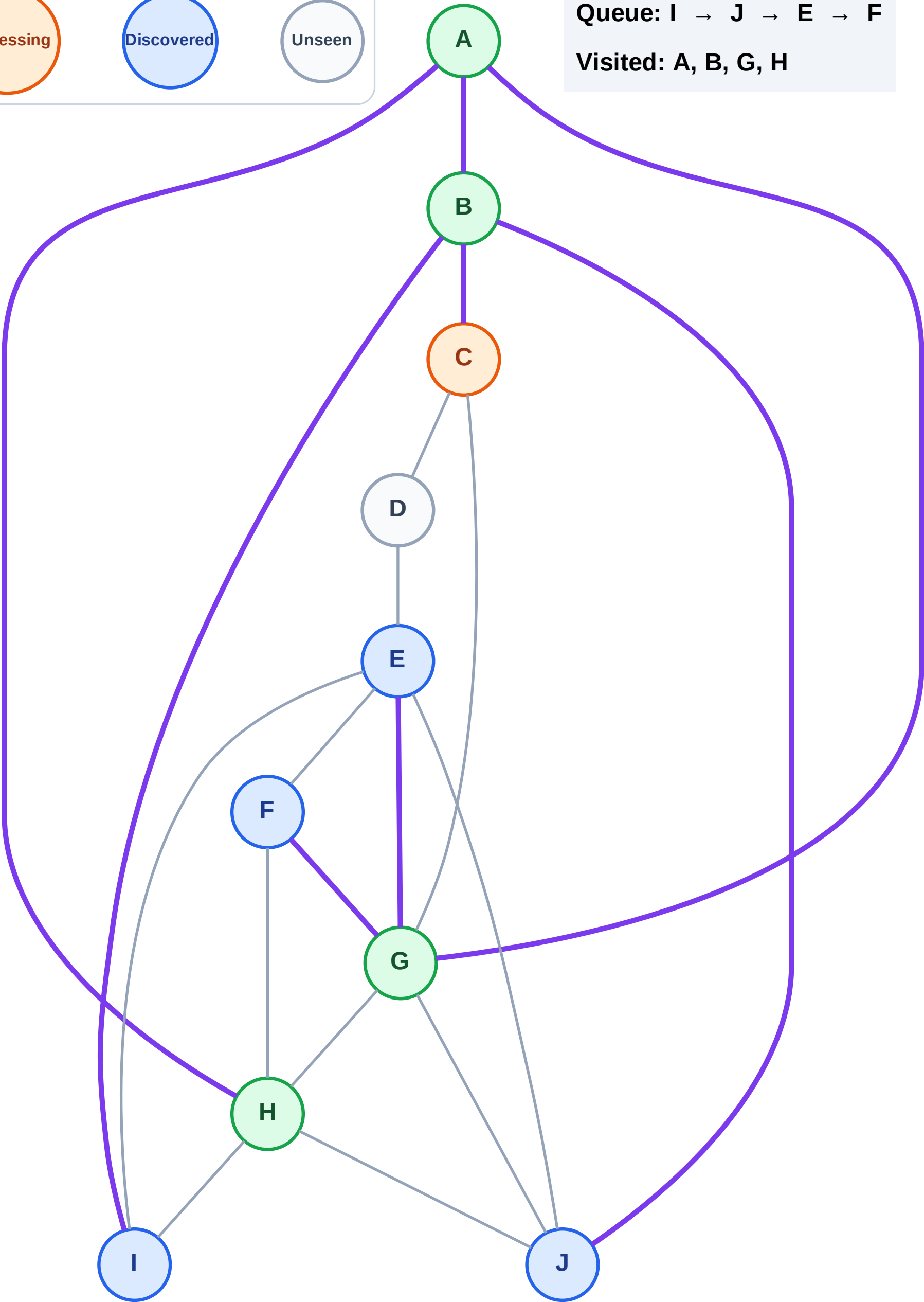
Processing

Discovered

Unseen

Queue: I → J → E → F

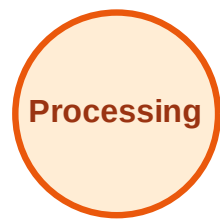
Visited: A, B, G, H



BFS Traversal

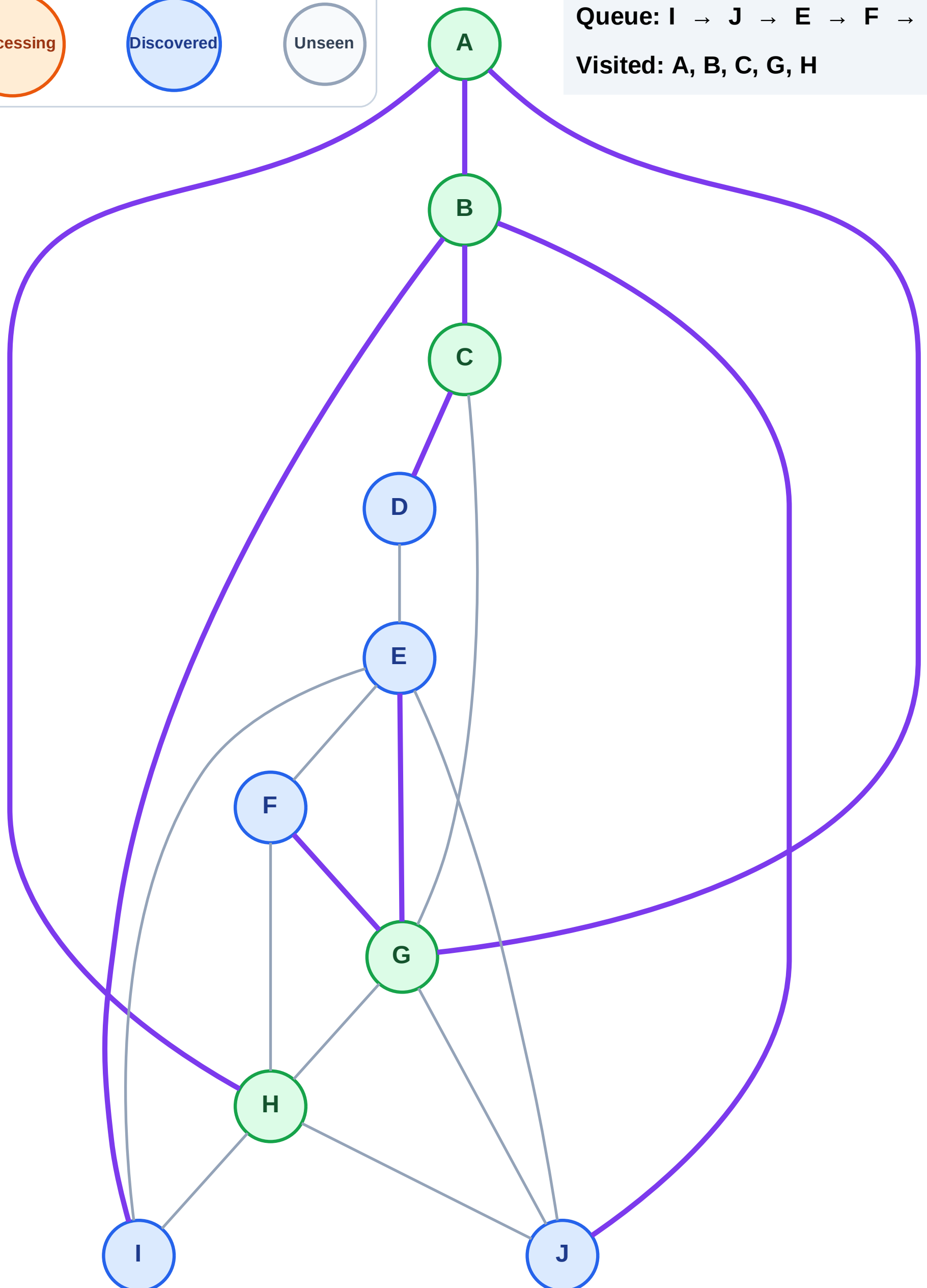
Step 11: Discover D from C

Legend



Queue: I → J → E → F → D

Visited: A, B, C, G, H



BFS Traversal

Step 12: Process node I

Legend

Complete

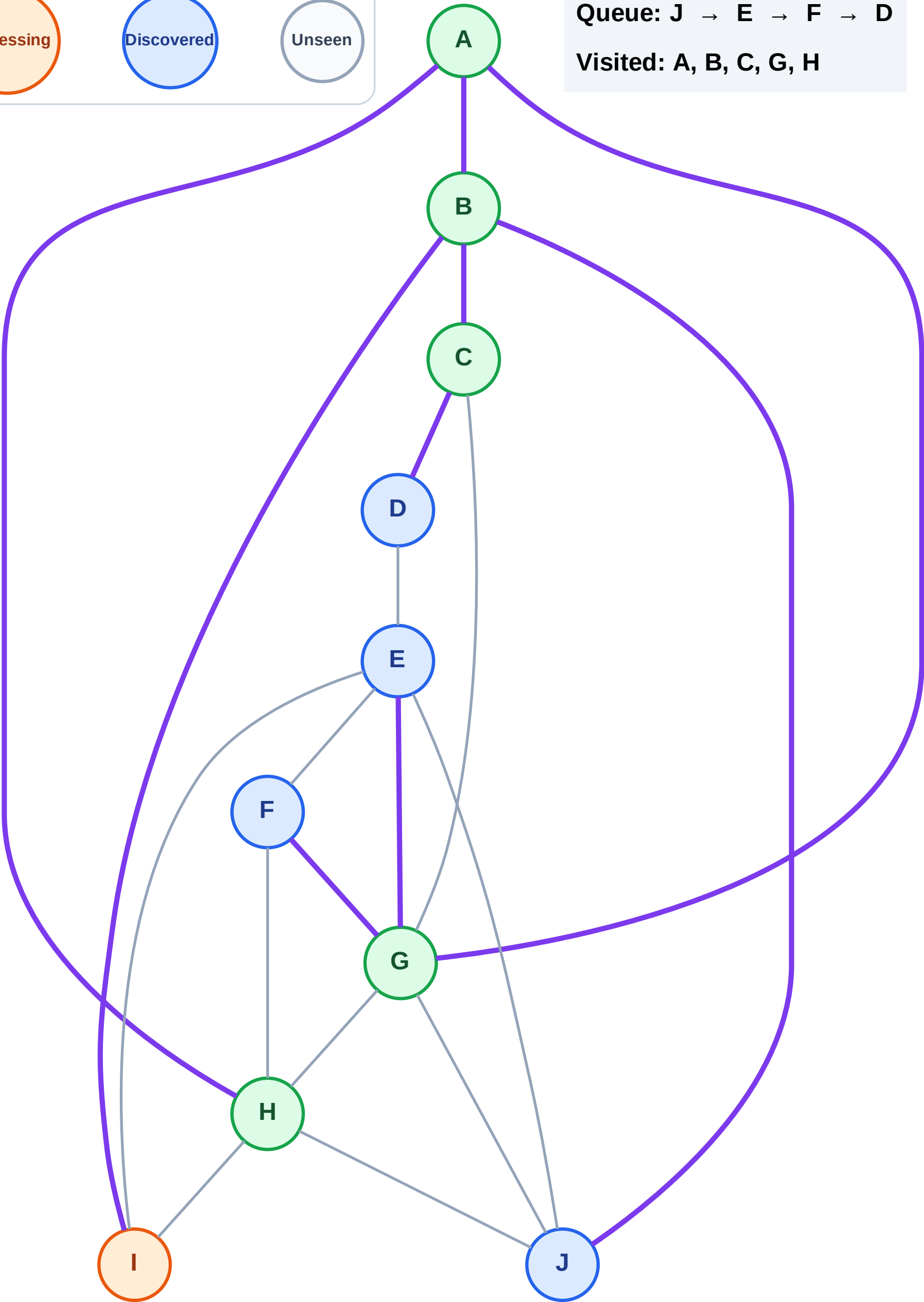
Processing

Discovered

Unseen

Queue: J → E → F → D

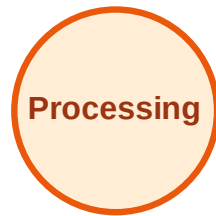
Visited: A, B, C, G, H



BFS Traversal

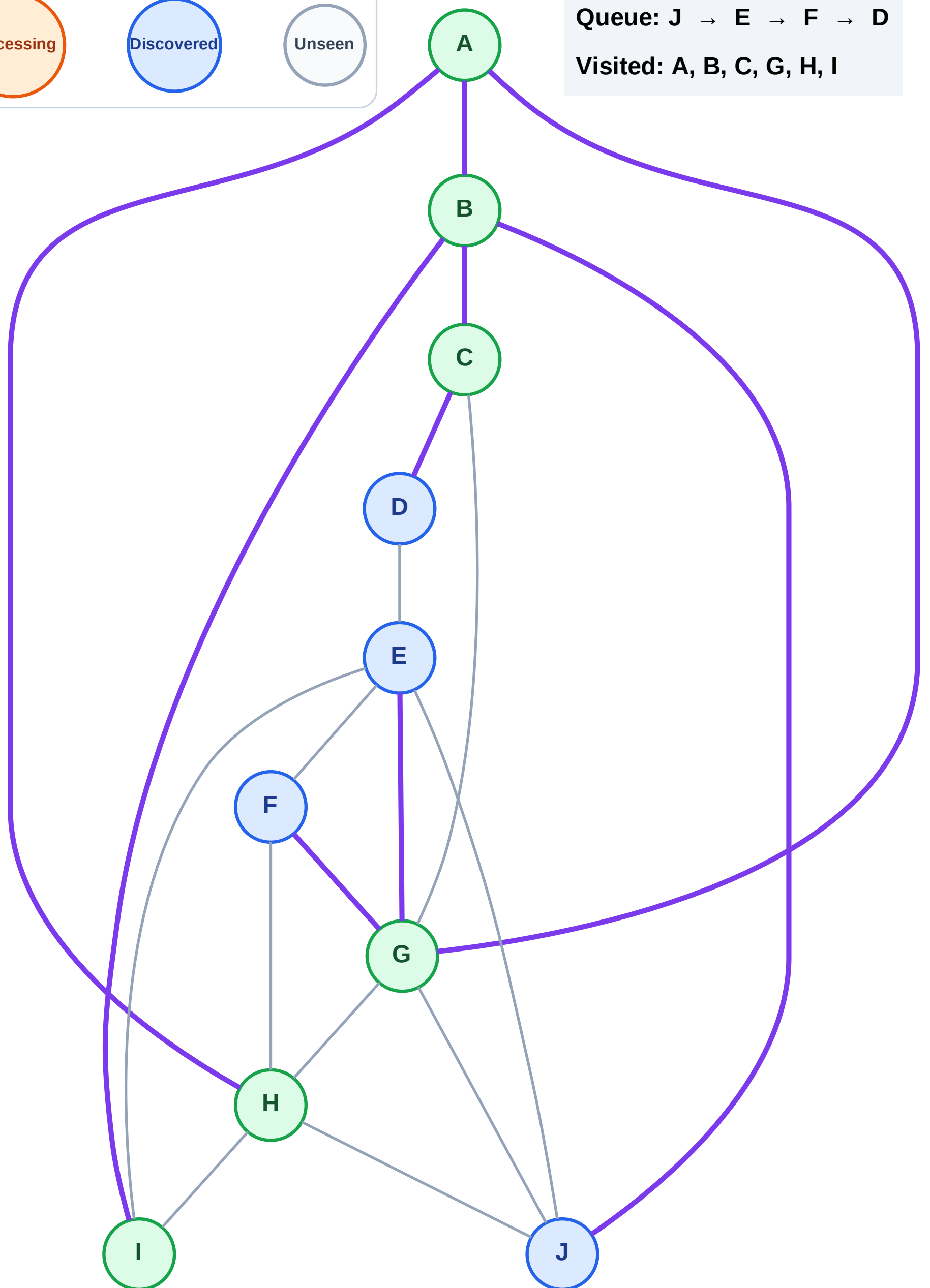
Step 13: No new neighbors from I

Legend



Queue: J → E → F → D

Visited: A, B, C, G, H, I



BFS Traversal

Step 14: Process node J

Legend

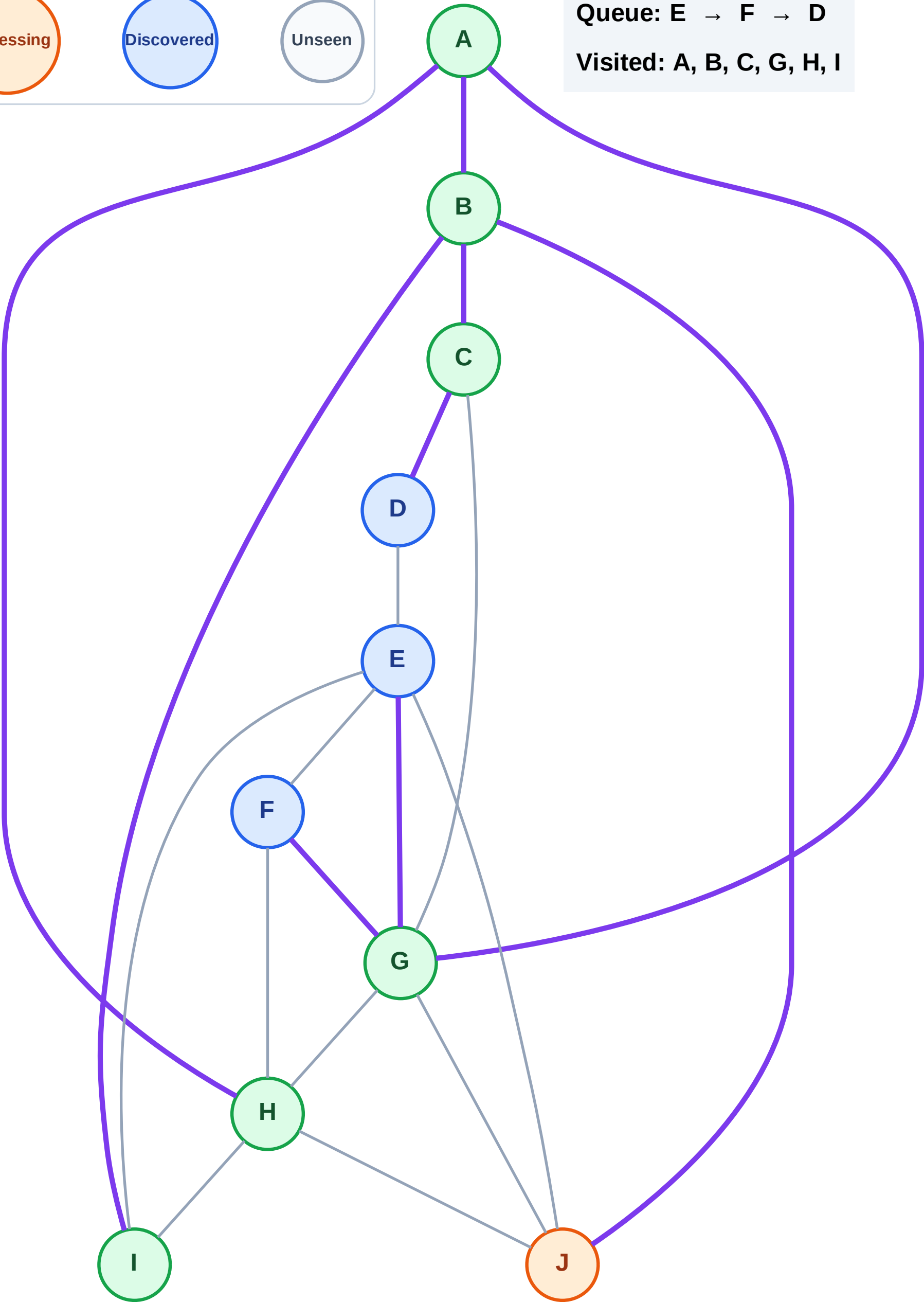
Complete

Processing

Discovered

Unseen

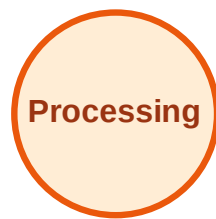
Queue: E → F → D
Visited: A, B, C, G, H, I



BFS Traversal

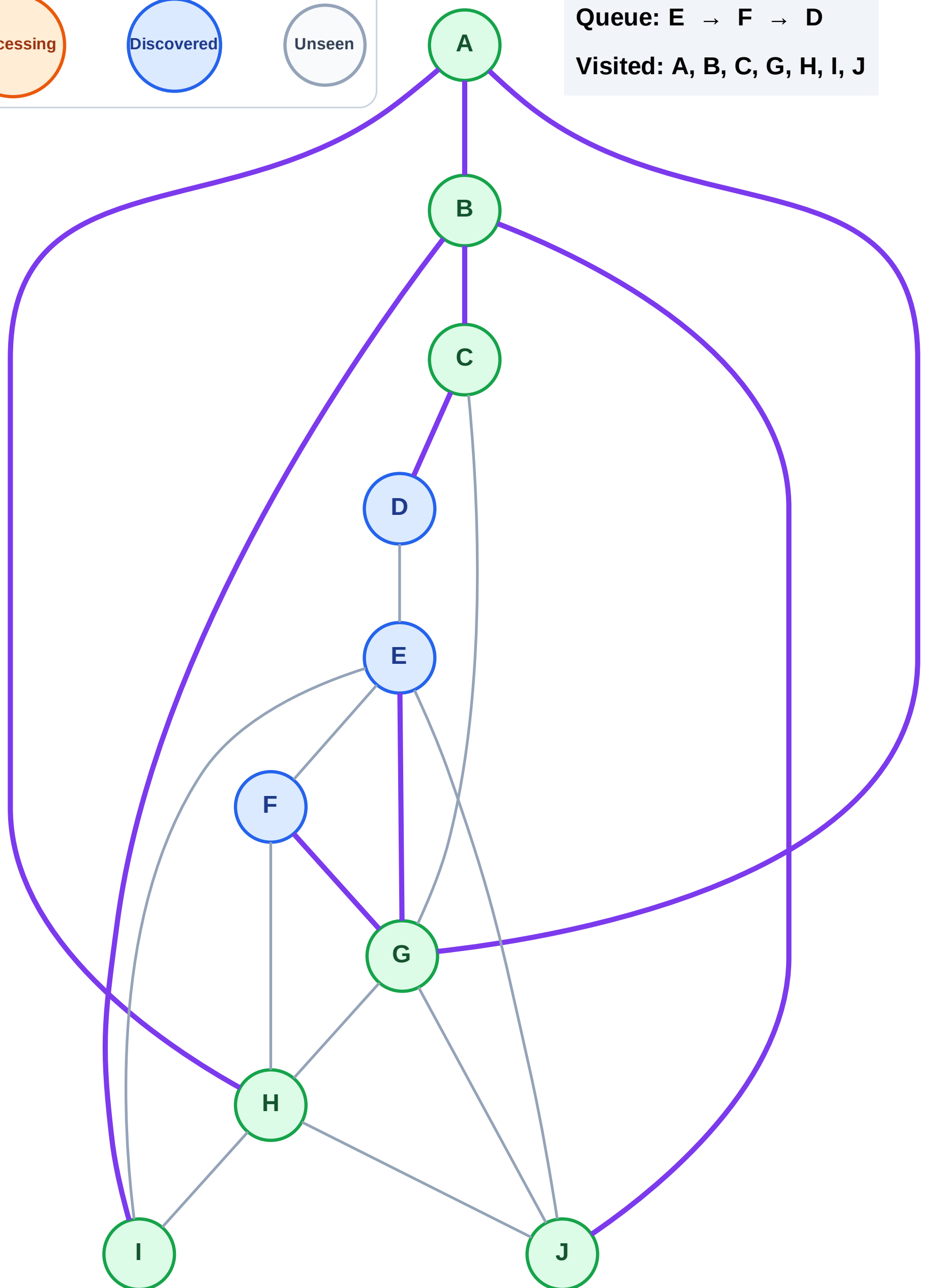
Step 15: No new neighbors from J

Legend



Queue: E → F → D

Visited: A, B, C, G, H, I, J



BFS Traversal

Step 16: Process node E

Legend

Complete

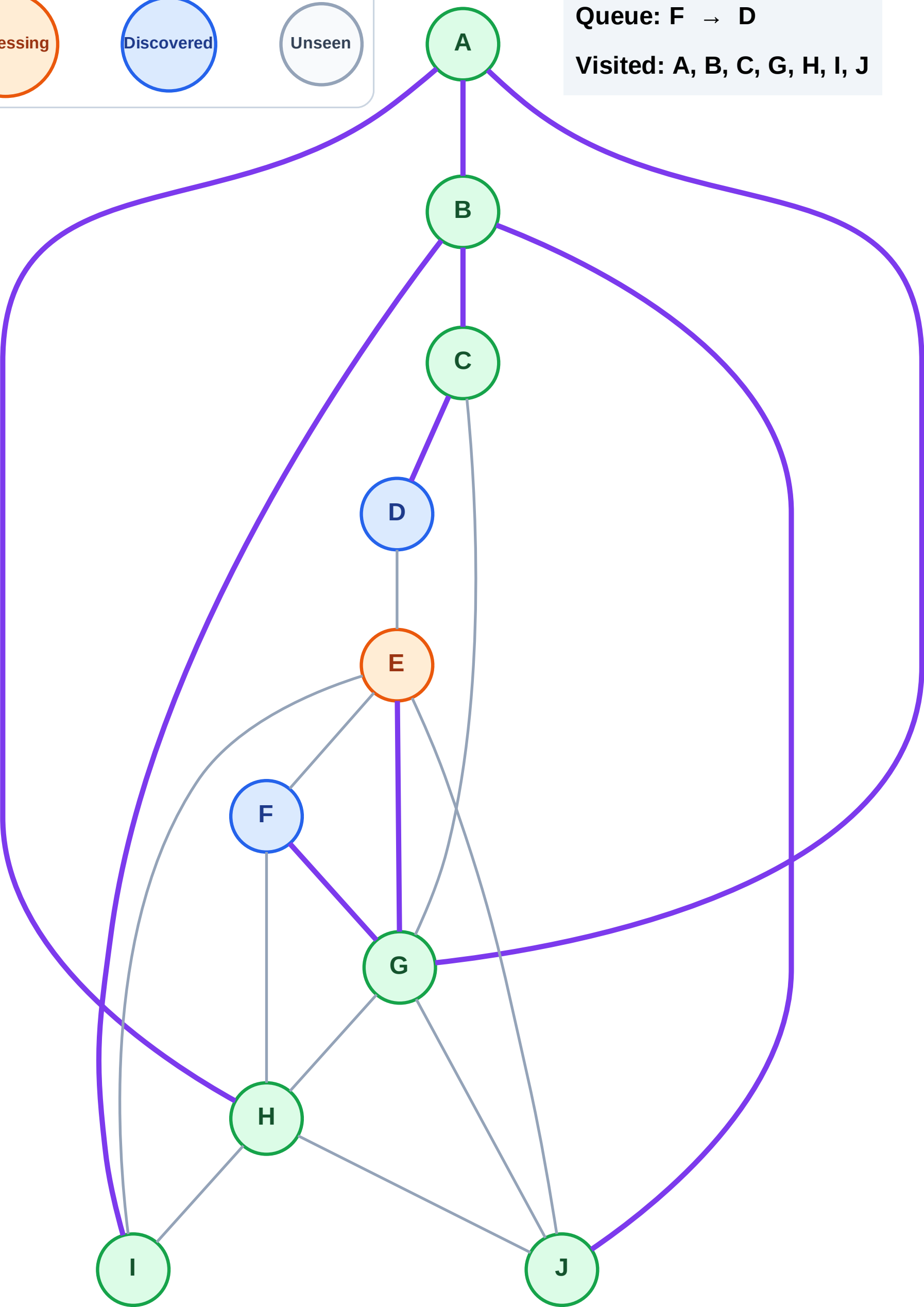
Processing

Discovered

Unseen

Queue: F → D

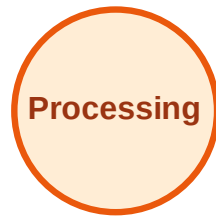
Visited: A, B, C, G, H, I, J



BFS Traversal

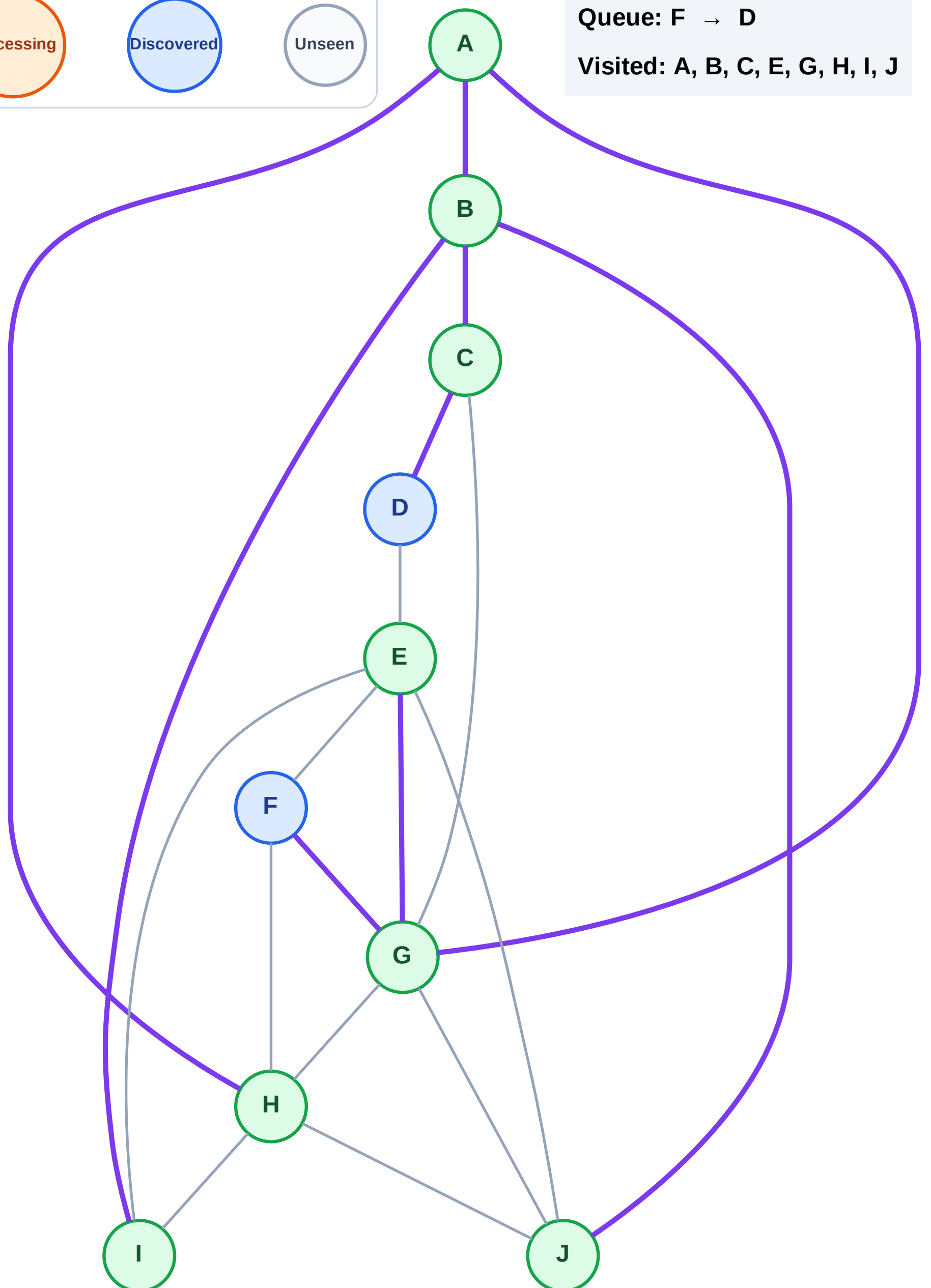
Step 17: No new neighbors from E

Legend



Queue: F → D

Visited: A, B, C, E, G, H, I, J



BFS Traversal

Step 18: Process node F

Legend

Complete

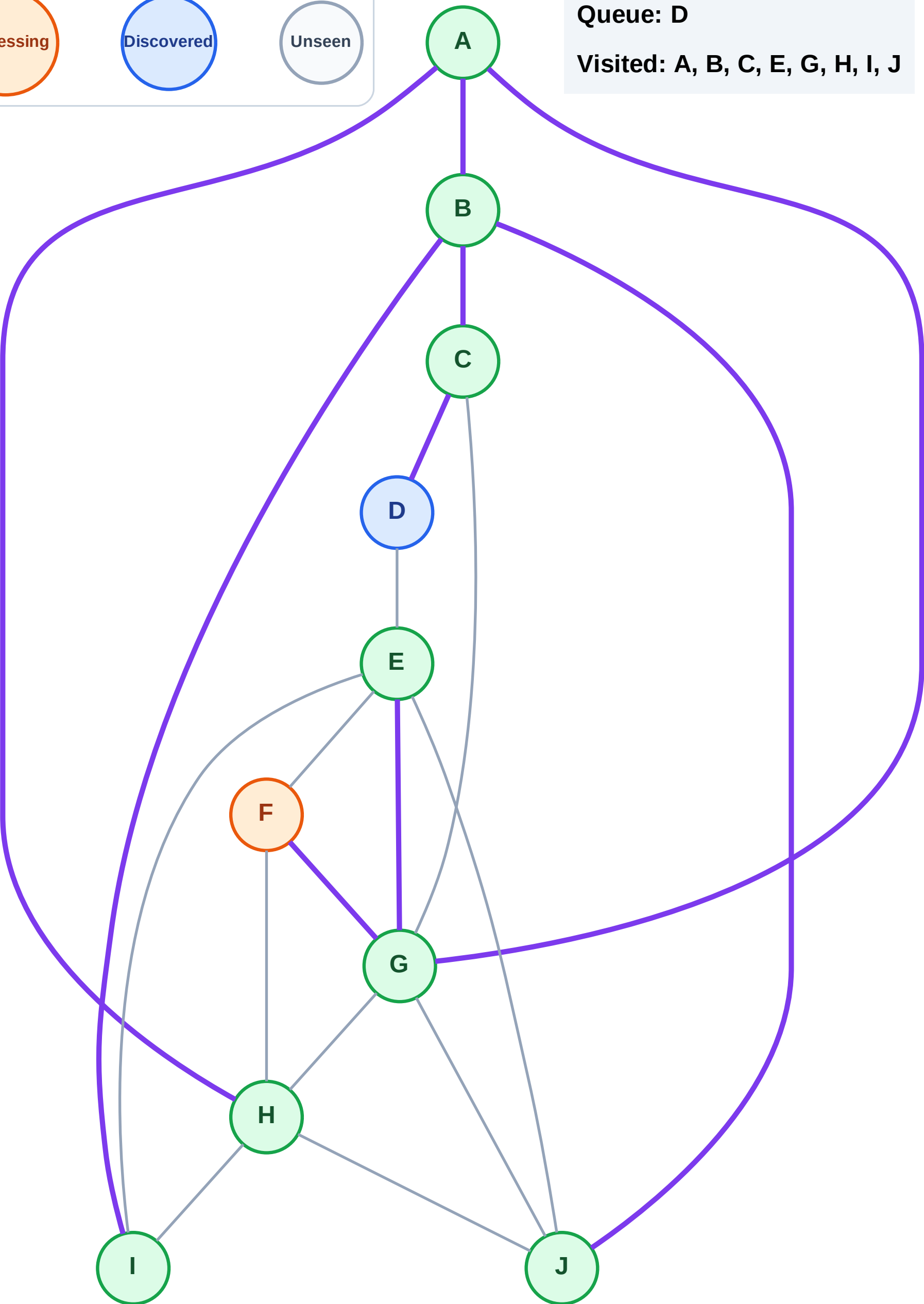
Processing

Discovered

Unseen

Queue: D

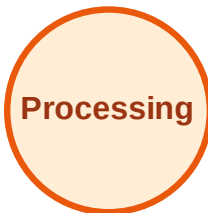
Visited: A, B, C, E, G, H, I, J



BFS Traversal

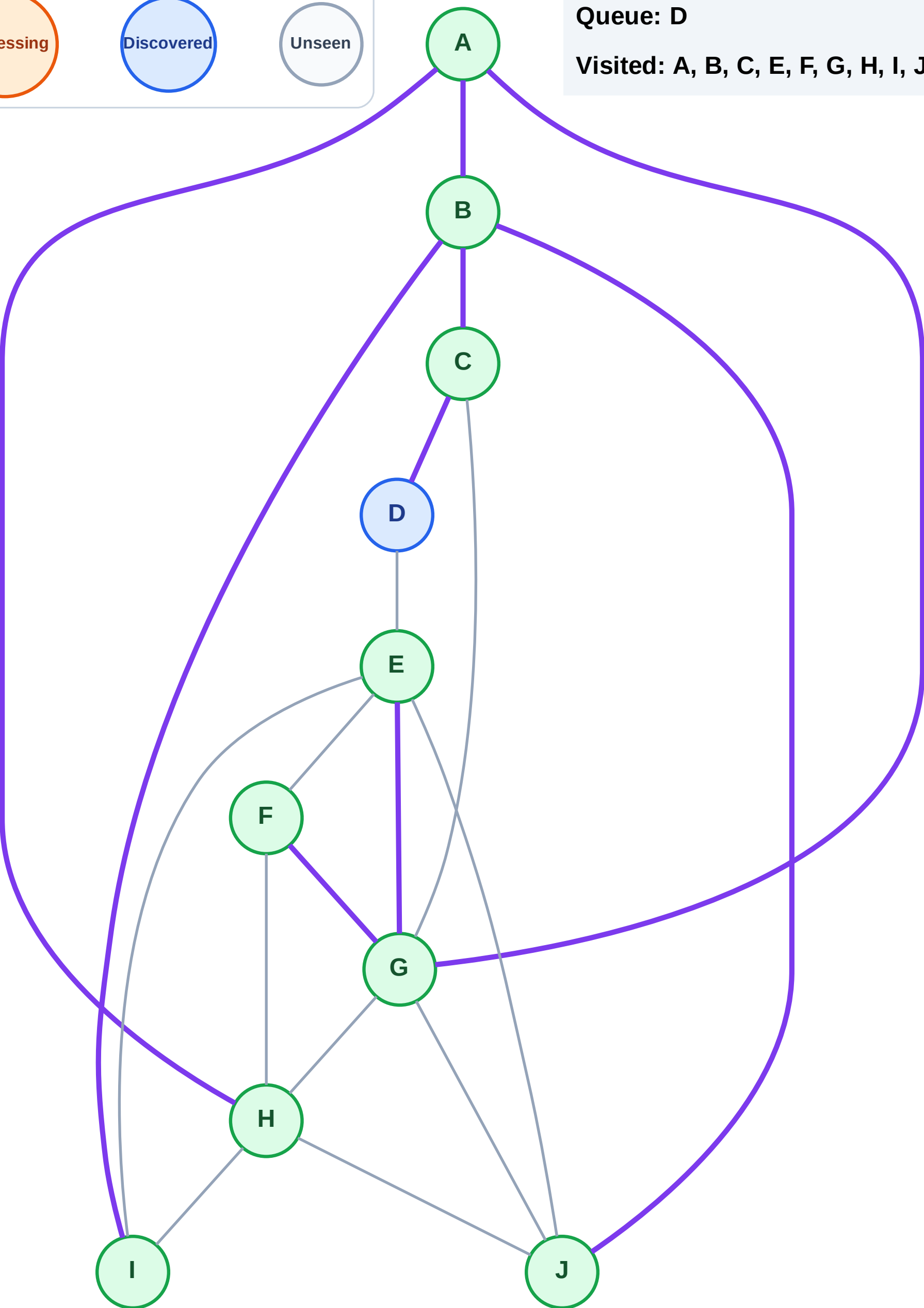
Step 19: No new neighbors from F

Legend



Queue: D

Visited: A, B, C, E, F, G, H, I, J



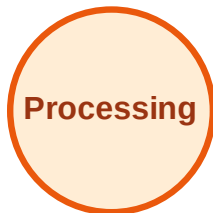
BFS Traversal

Step 20: Process node D

Legend



Complete



Processing



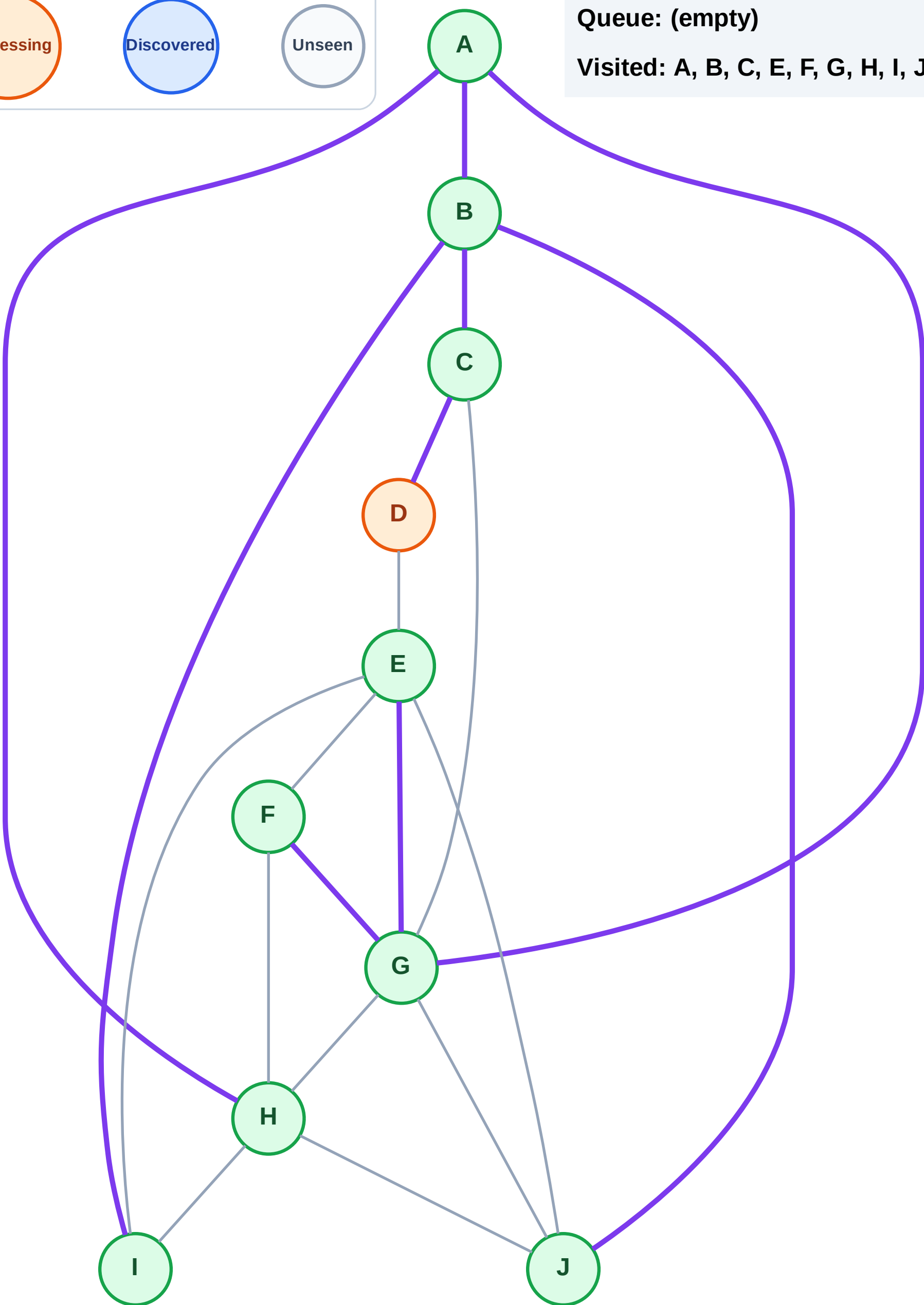
Discovered



Unseen

Queue: (empty)

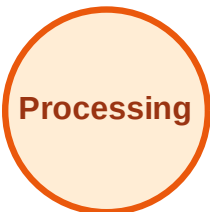
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

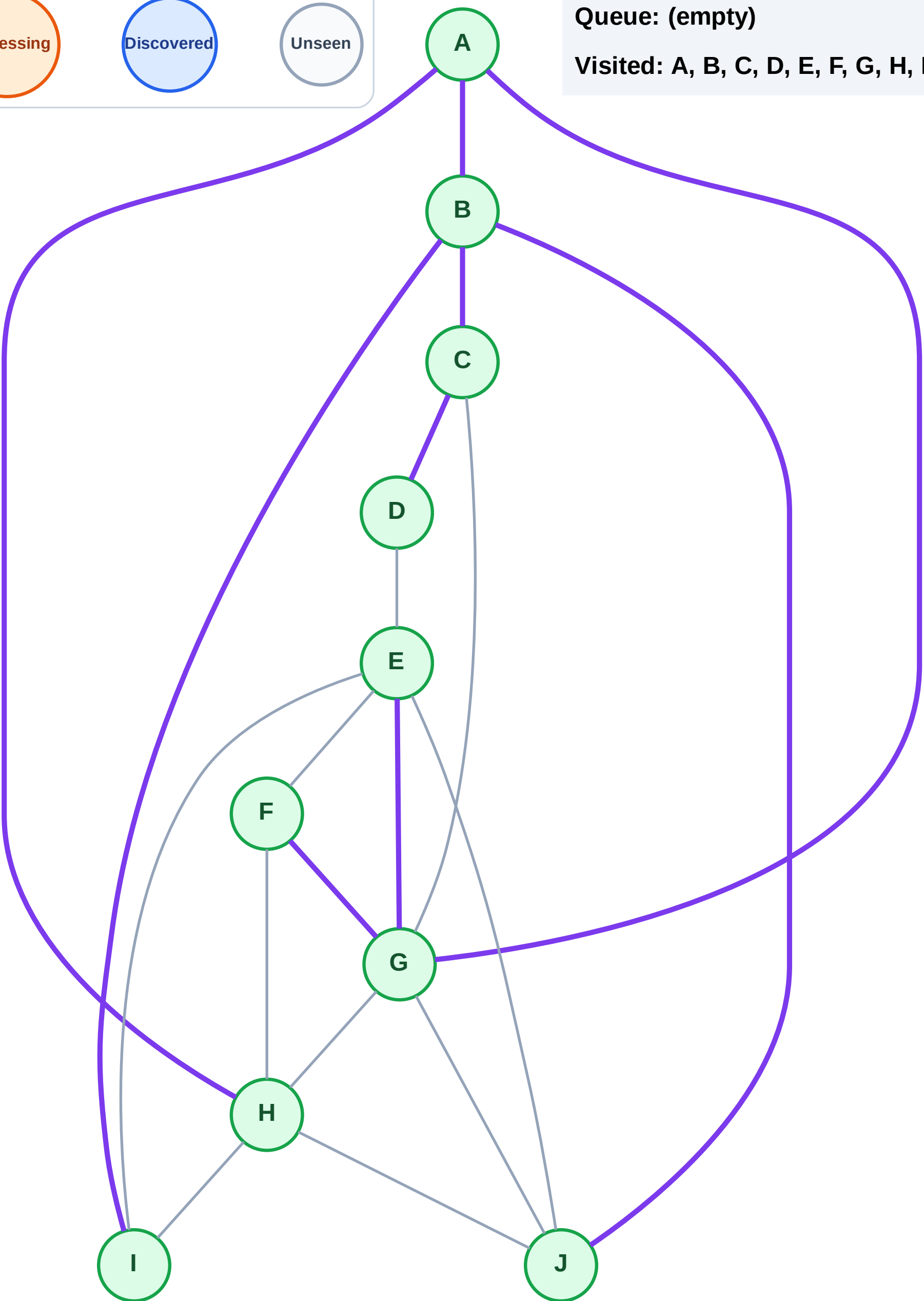
Step 21: No new neighbors from D

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at A

Legend

Complete

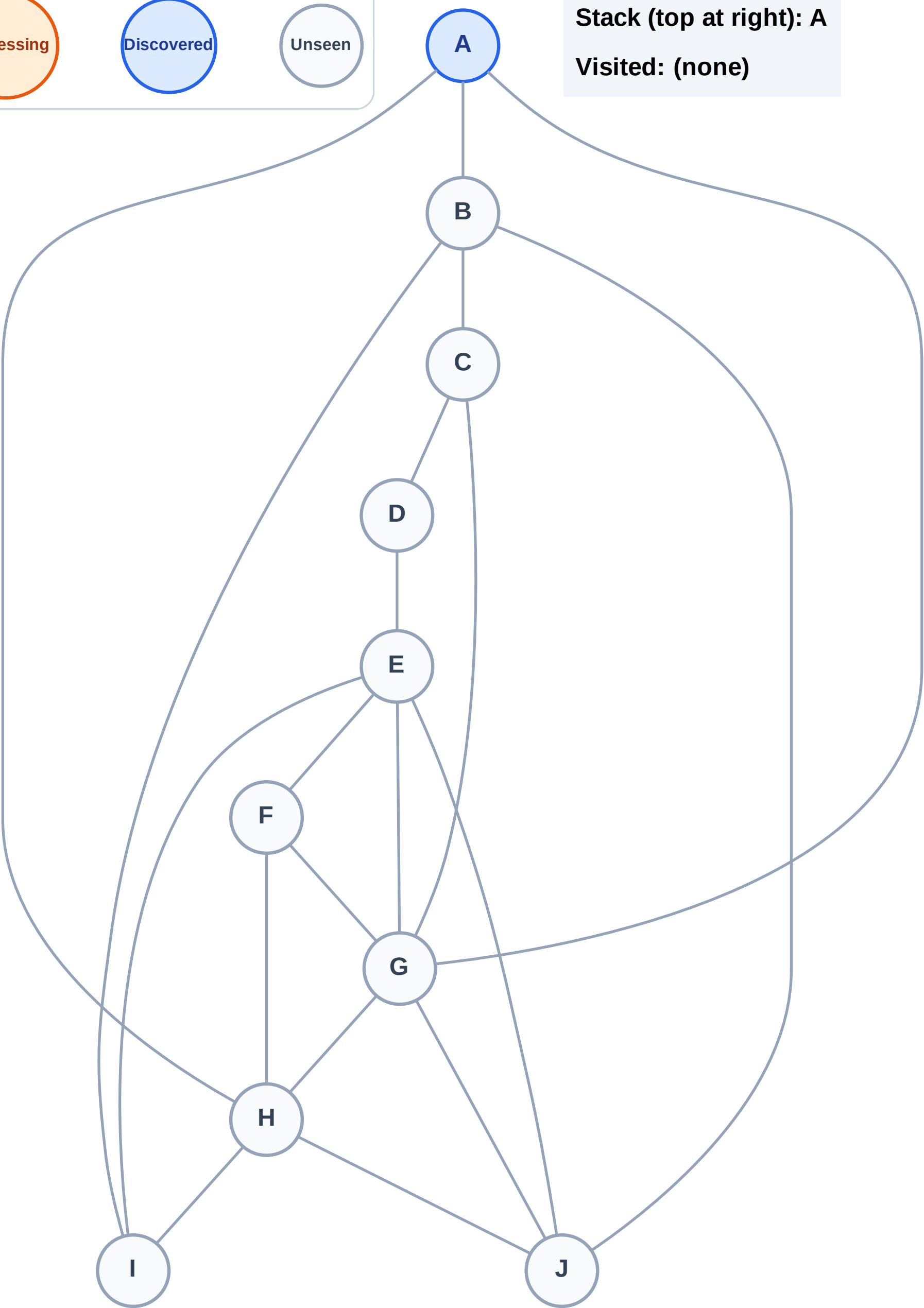
Processing

Discovered

Unseen

Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

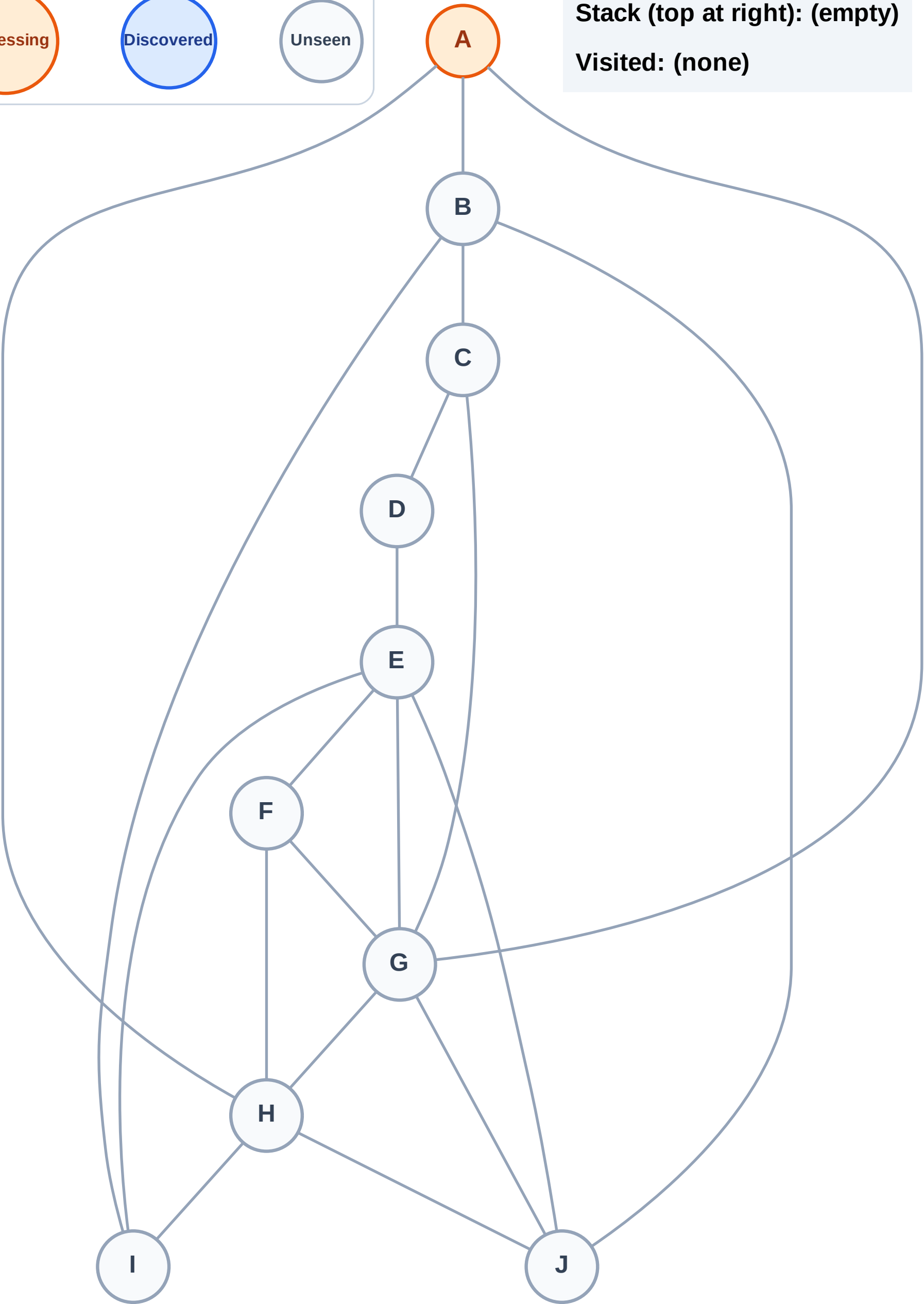
Complete

Processing

Discovered

Unseen

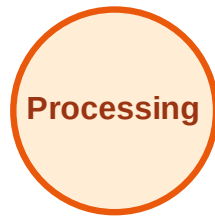
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

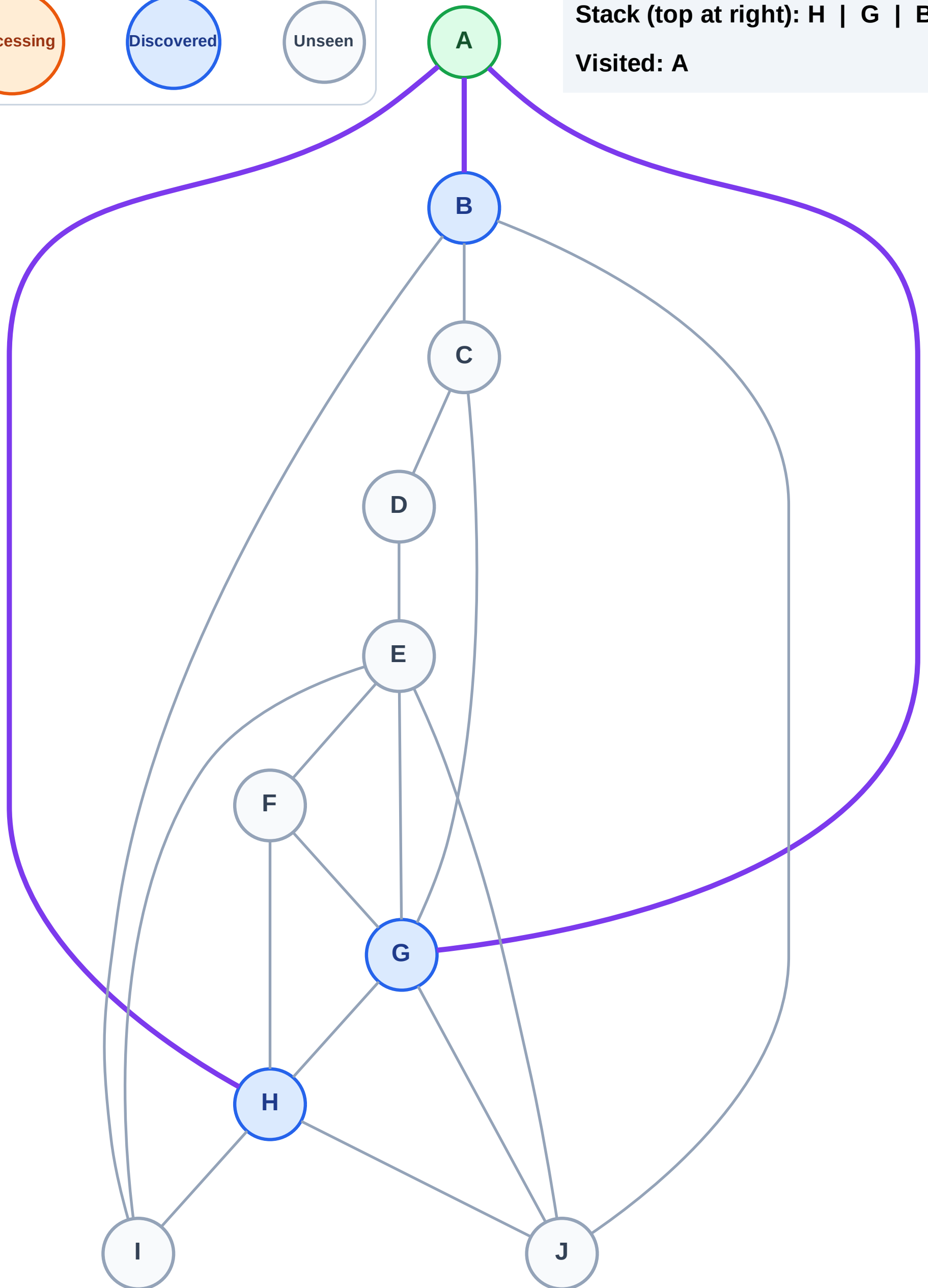
Step 3: Discover H, G, B from A

Legend



Stack (top at right): H | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

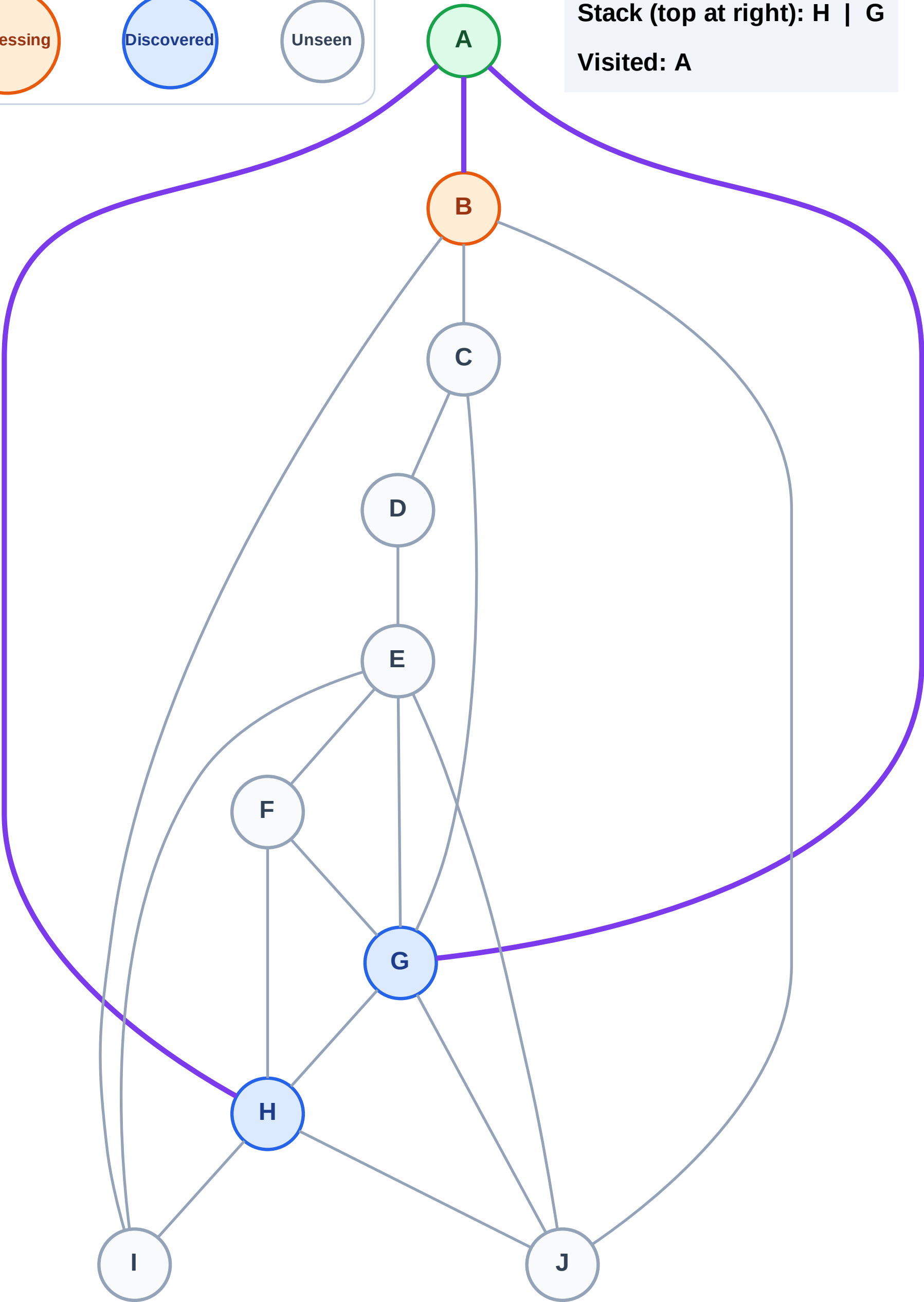
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A

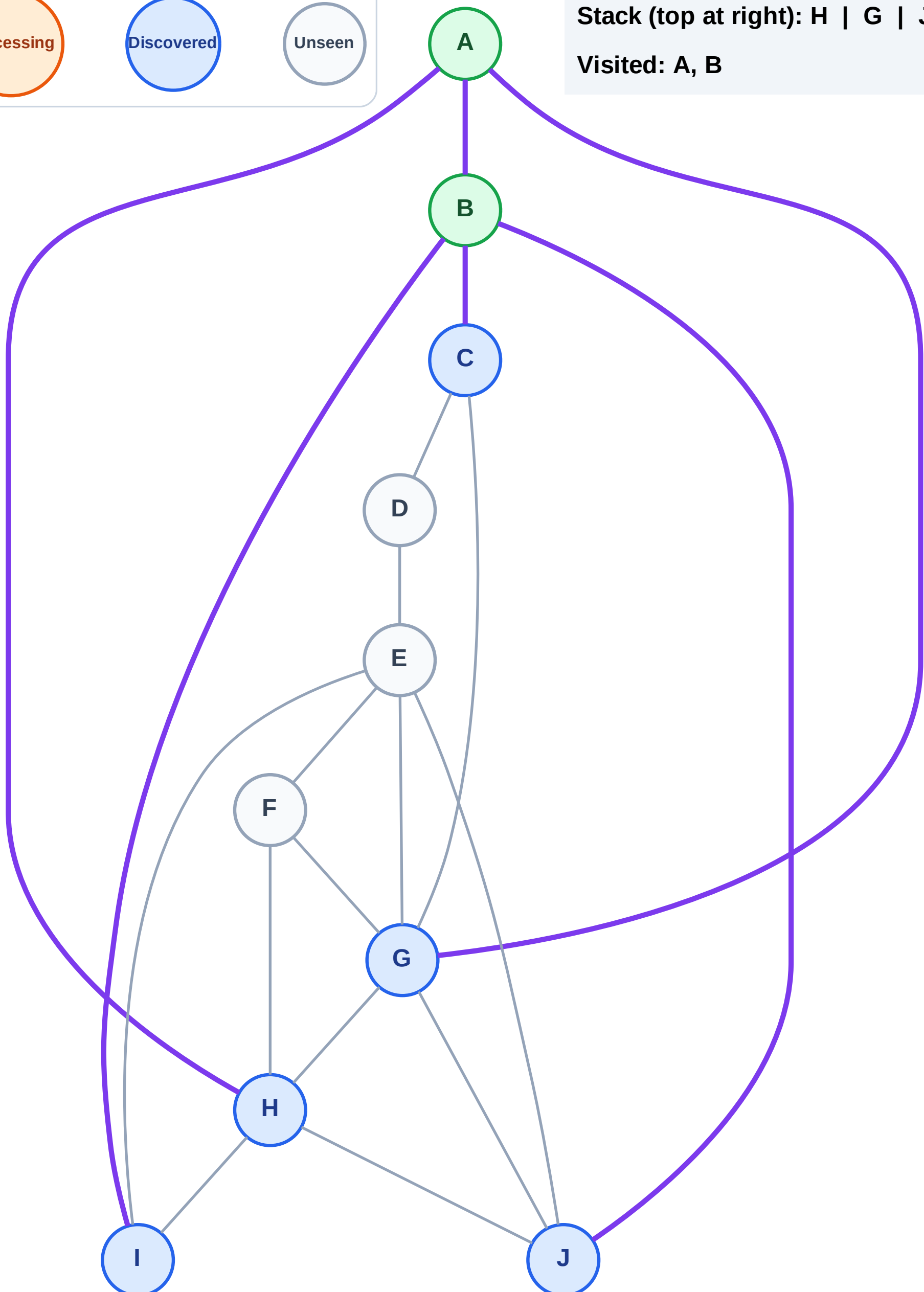


Step 5: Discover J, I, C from B

Legend

Unseen

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

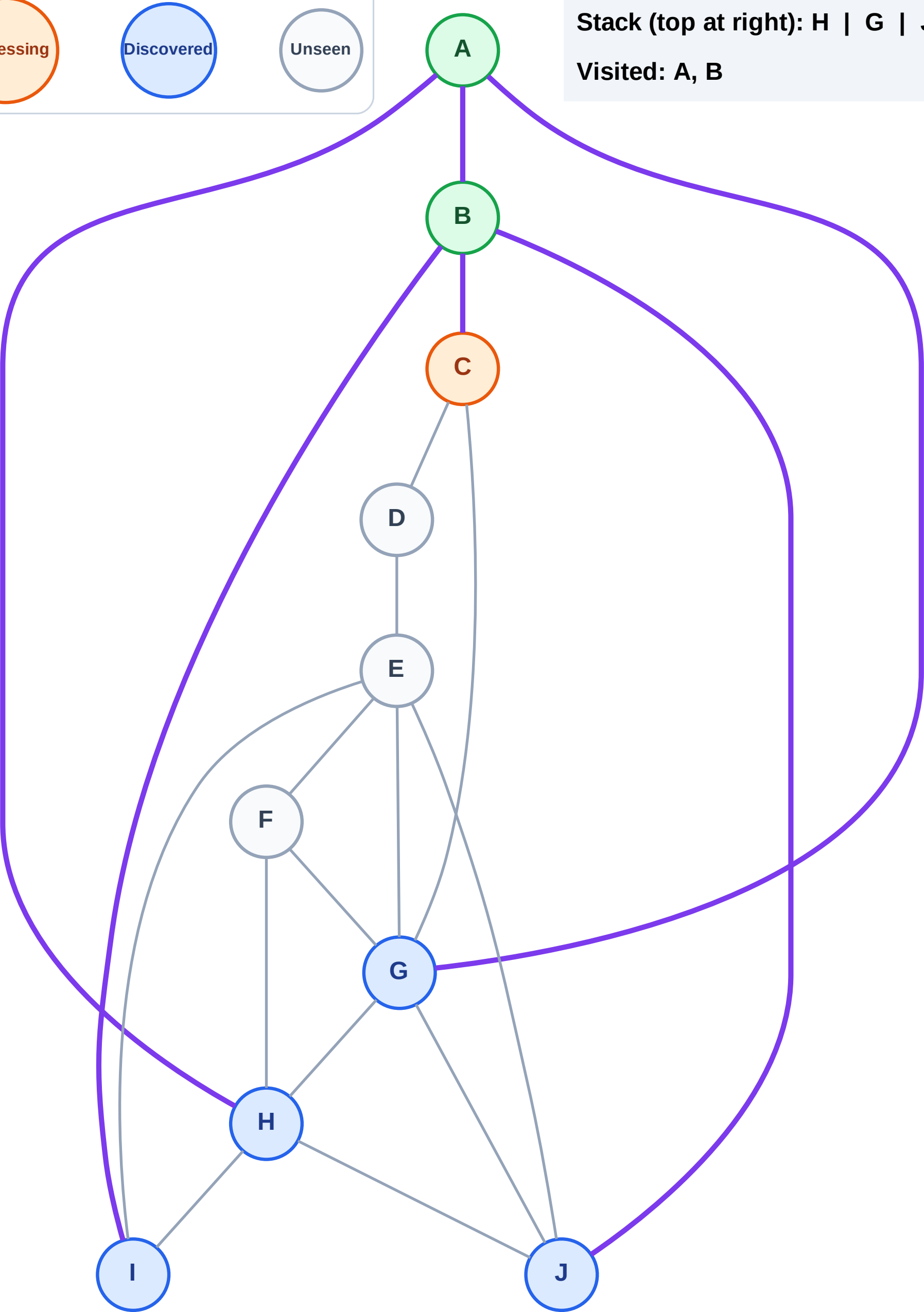
Processing

Discovered

Unseen

Stack (top at right): H | G | J | I

Visited: A, B

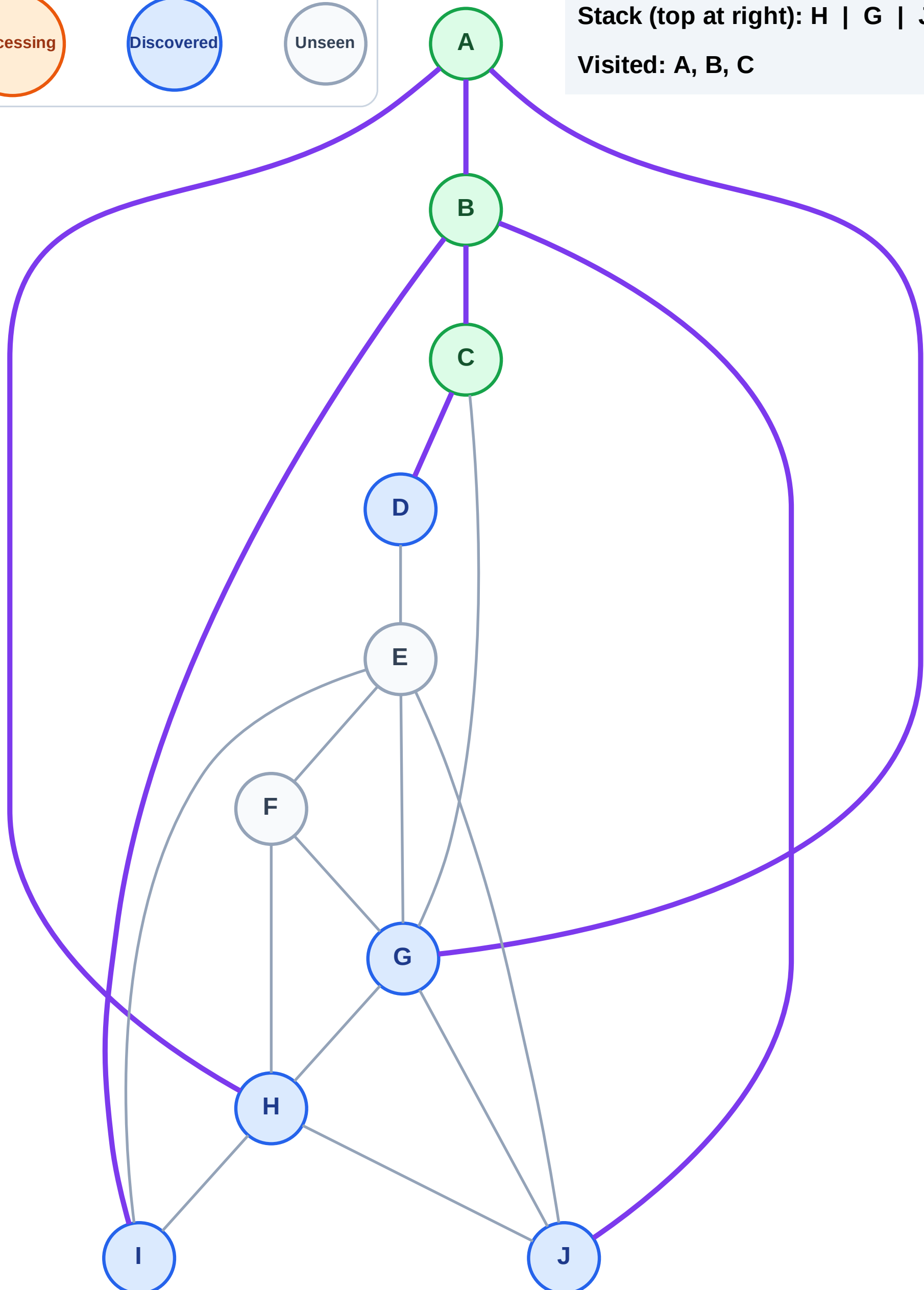


Step 7: Discover D from C

Legend

Unseen

Visited: A, B, C



DFS Traversal

Step 8: Process node D

Legend

Complete

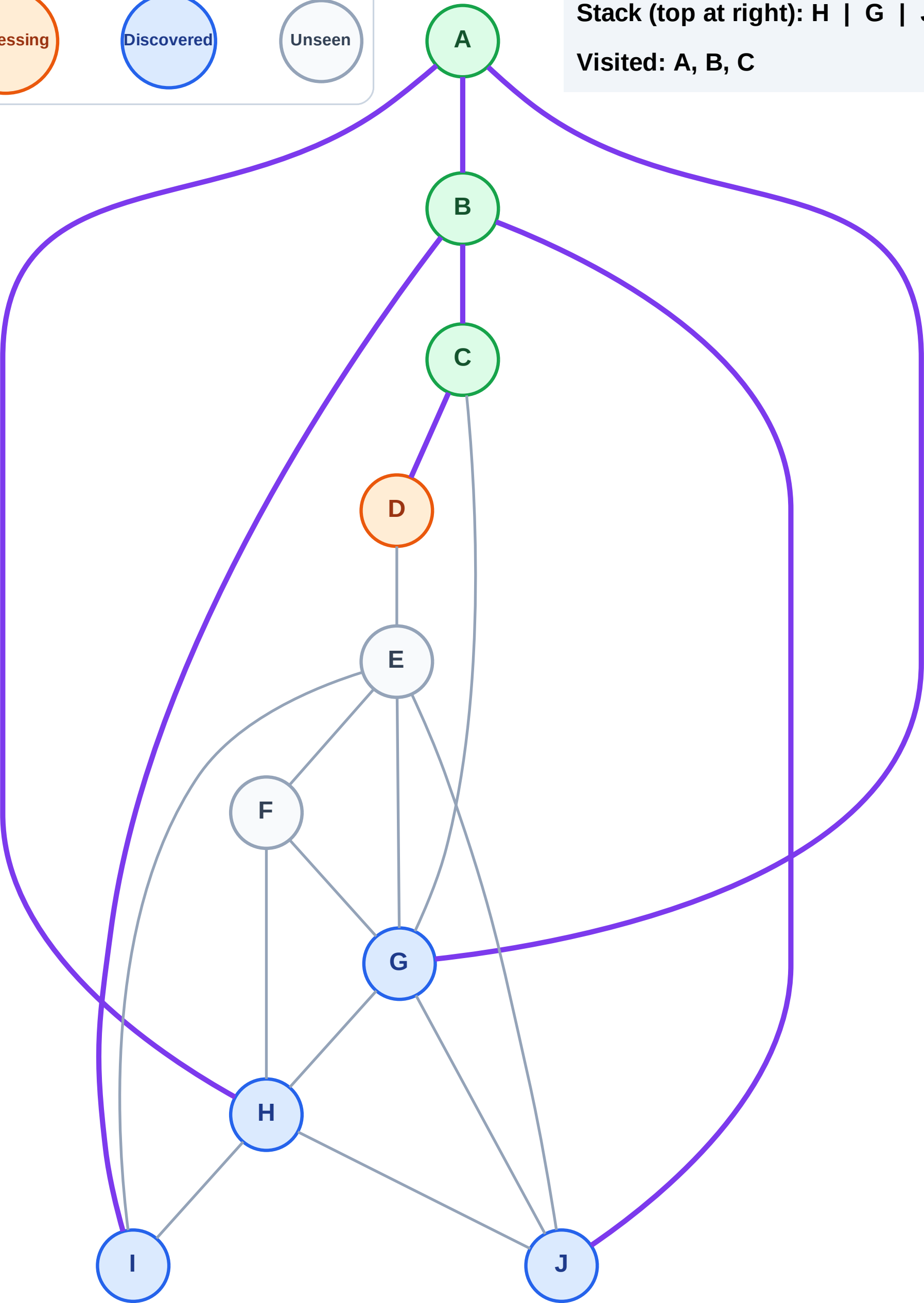
Processing

Discovered

Unseen

Stack (top at right): H | G | J | I

Visited: A, B, C

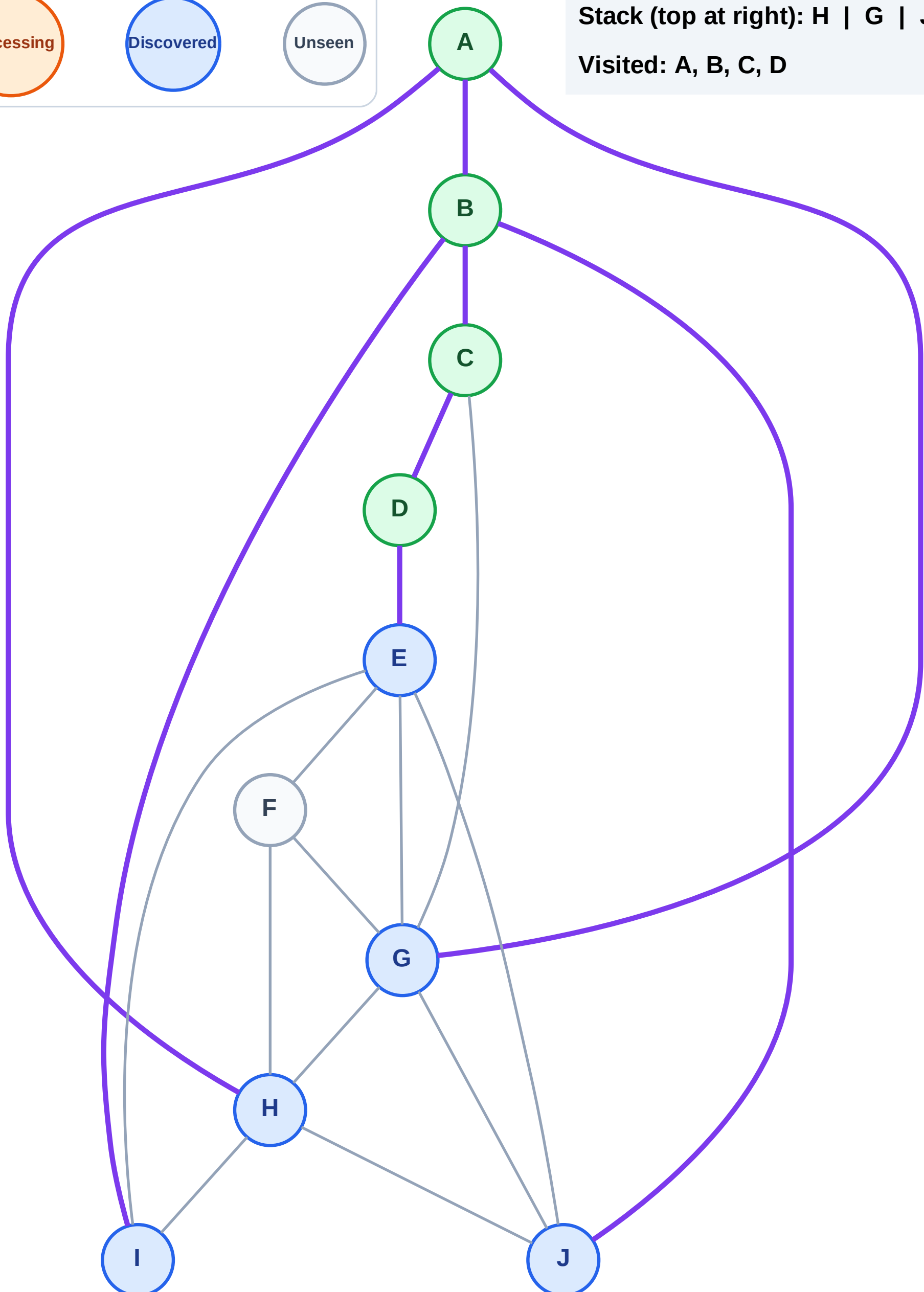


Step 9: Discover E from D

Legend

Unseen

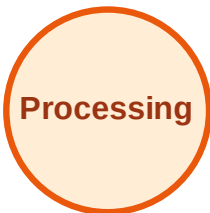
Visited: A, B, C, D



DFS Traversal

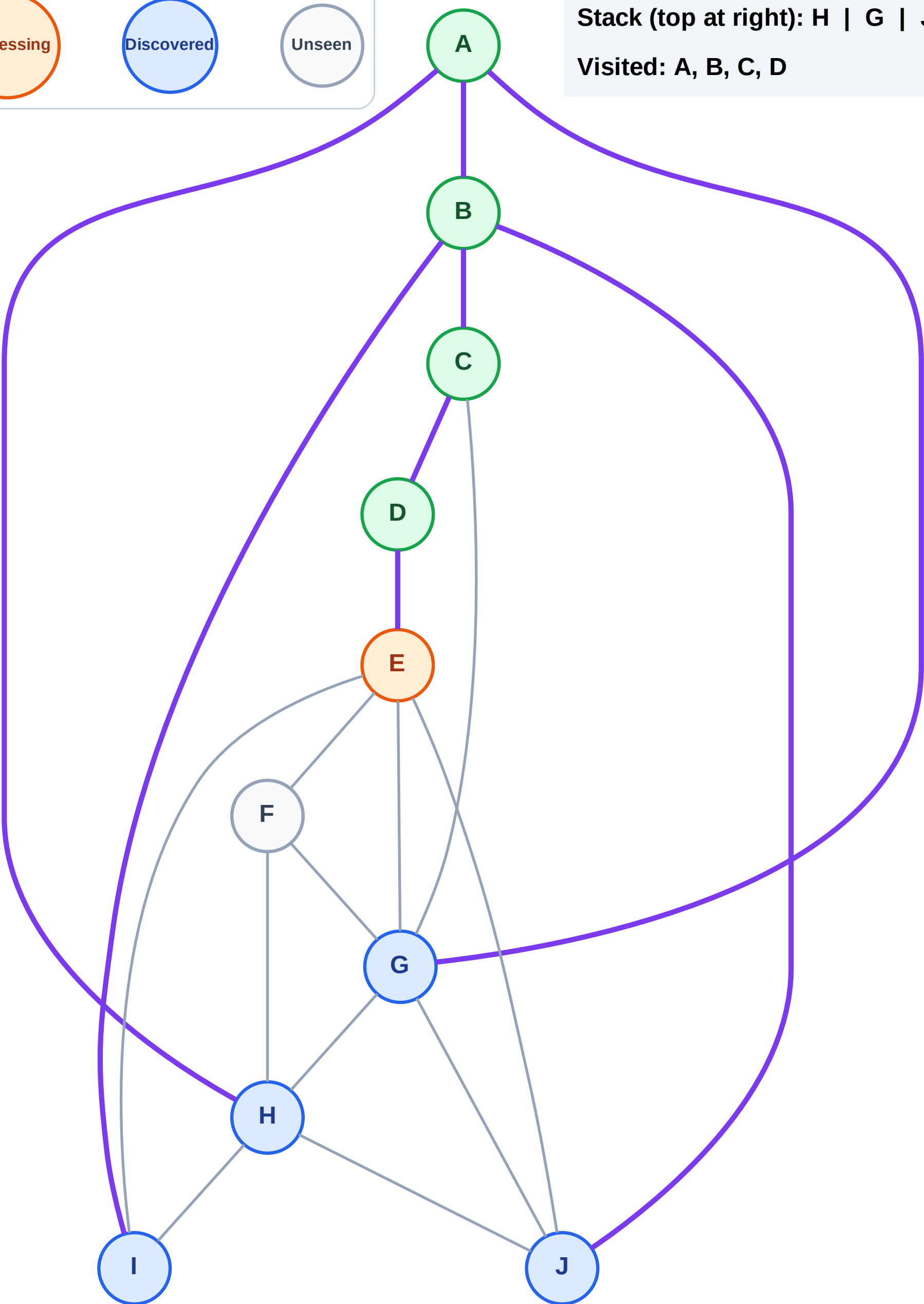
Step 10: Process node E

Legend



Stack (top at right): H | G | J | I

Visited: A, B, C, D



Step 11: Discover F from E

Legend

Unseen

Visited: A, B, C, D, E

DFS Traversal

Step 12: Process node F

Legend

Complete

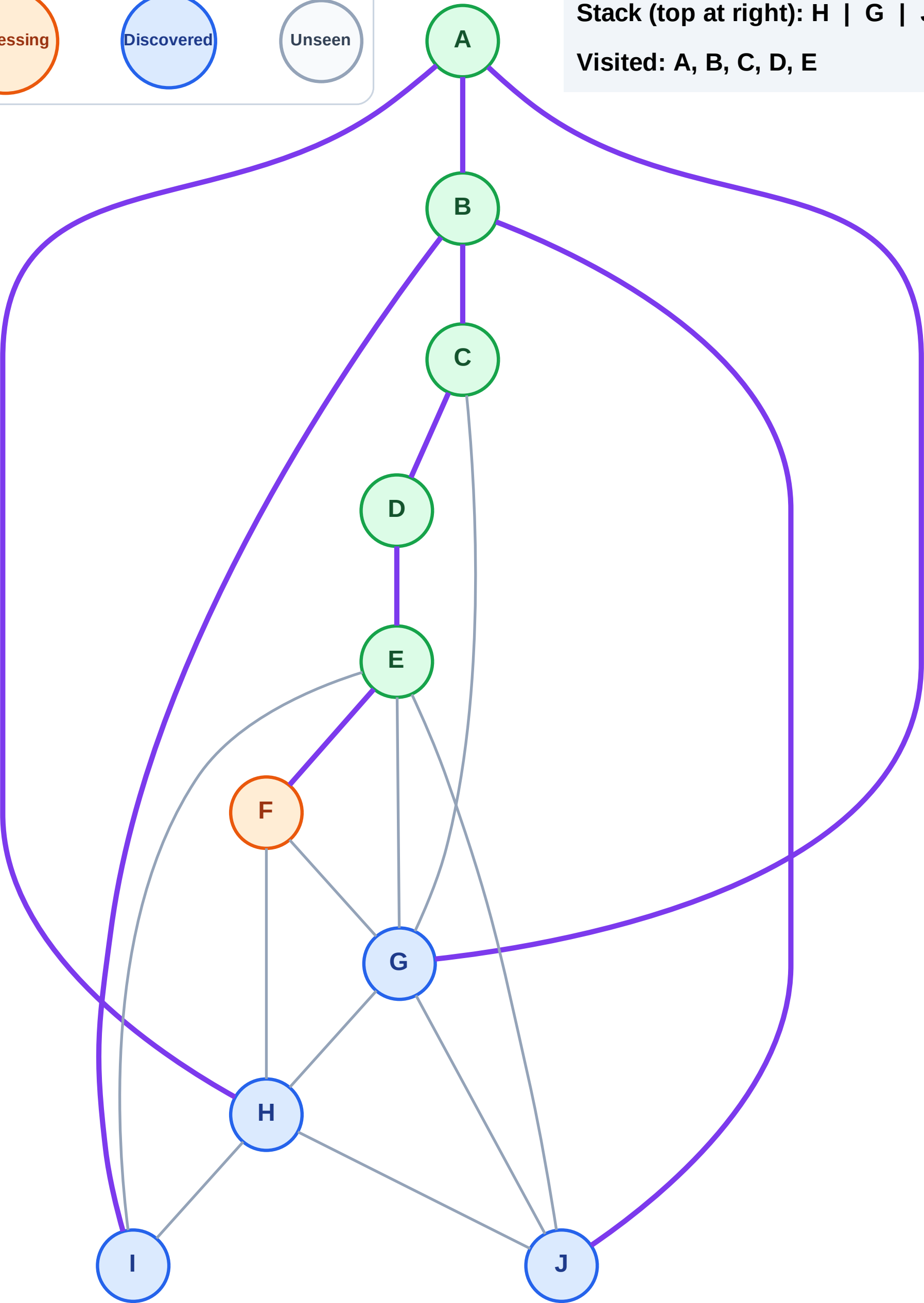
Processing

Discovered

Unseen

Stack (top at right): H | G | J | I

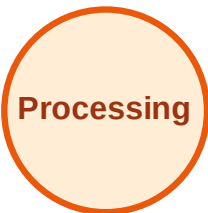
Visited: A, B, C, D, E



DFS Traversal

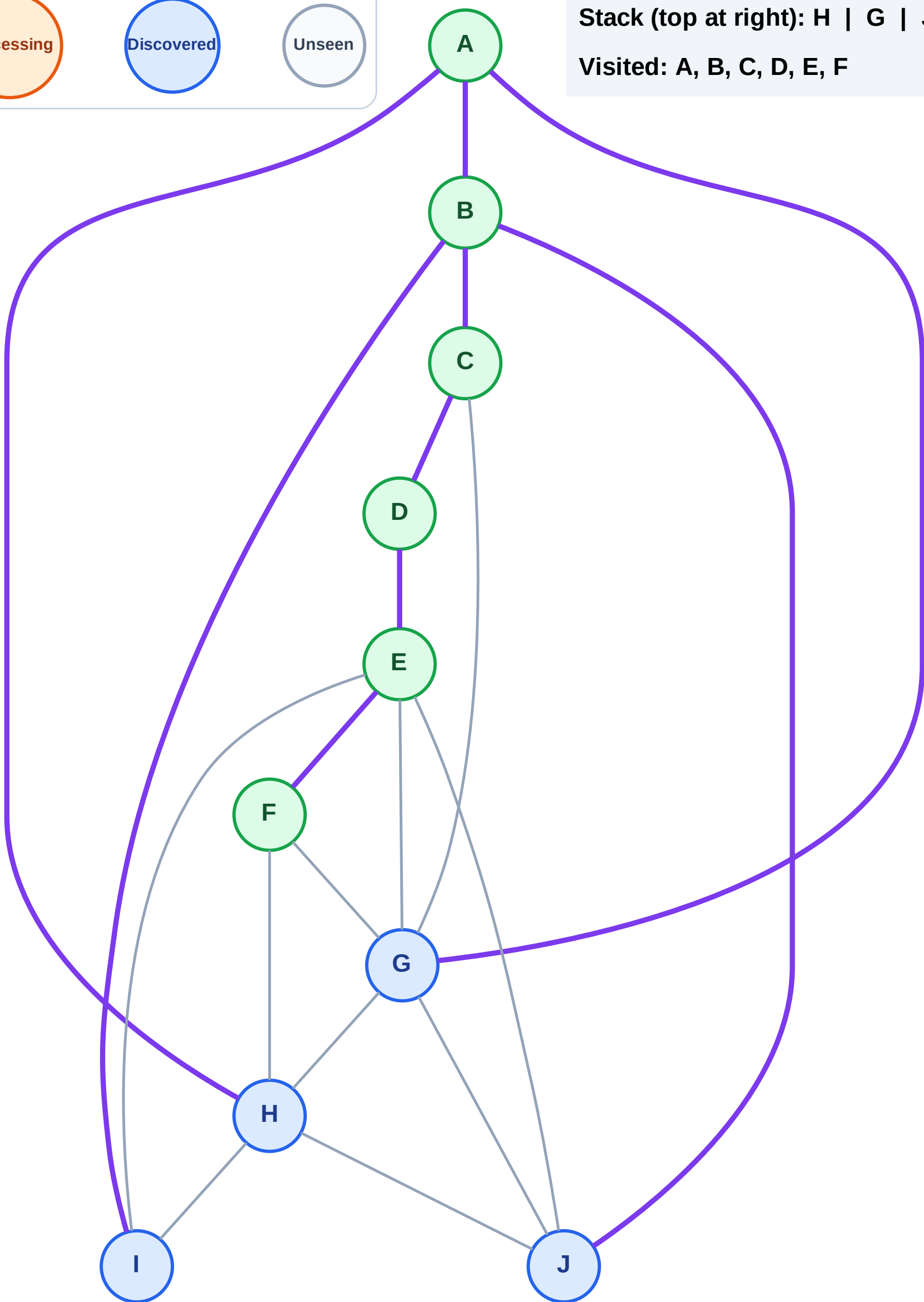
Step 13: No new neighbors from F

Legend



Stack (top at right): H | G | J | I

Visited: A, B, C, D, E, F

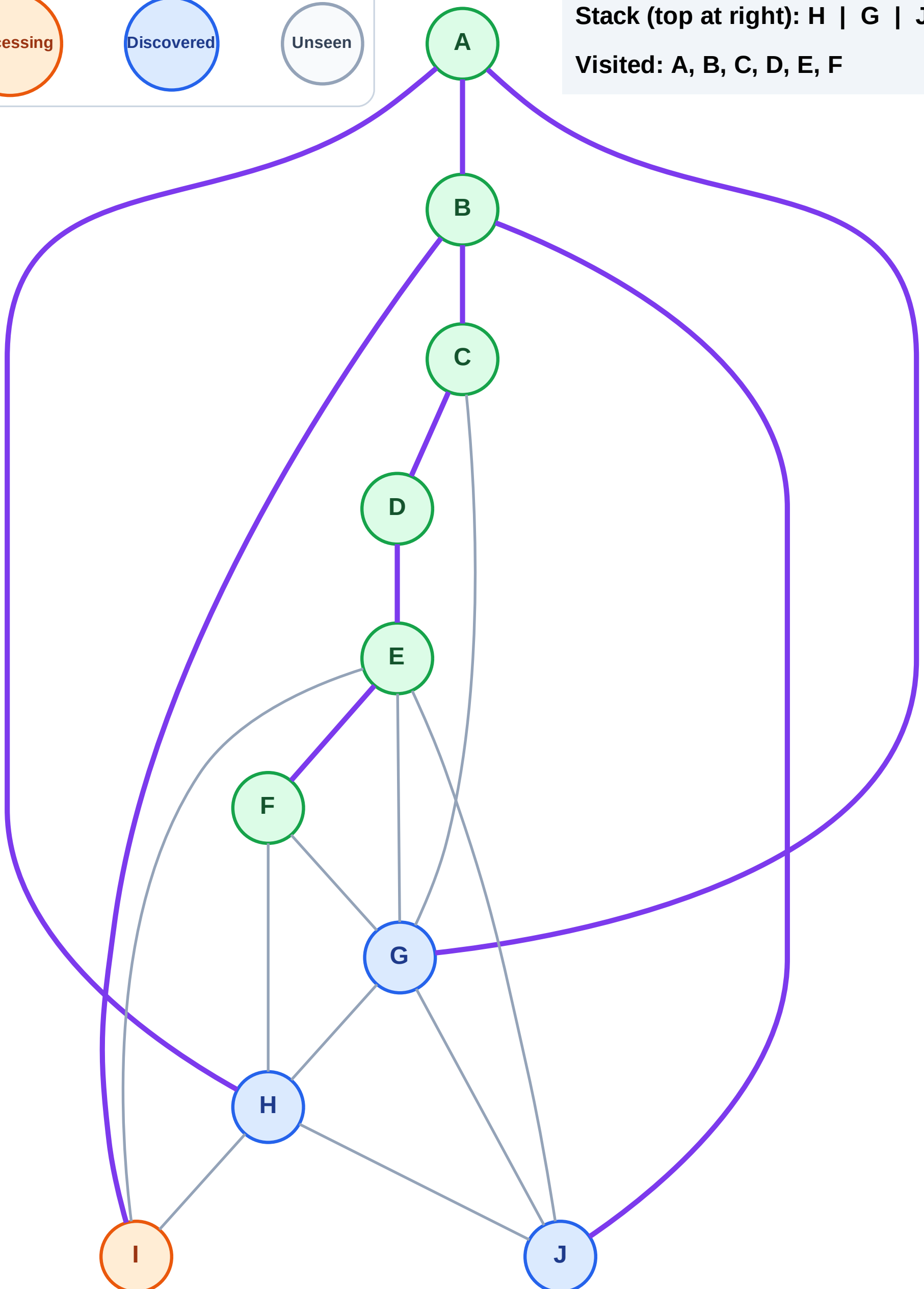


Step 14: Process node I

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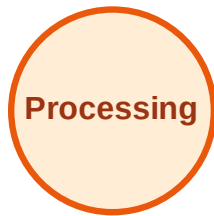
Visited: A B C D E E



DFS Traversal

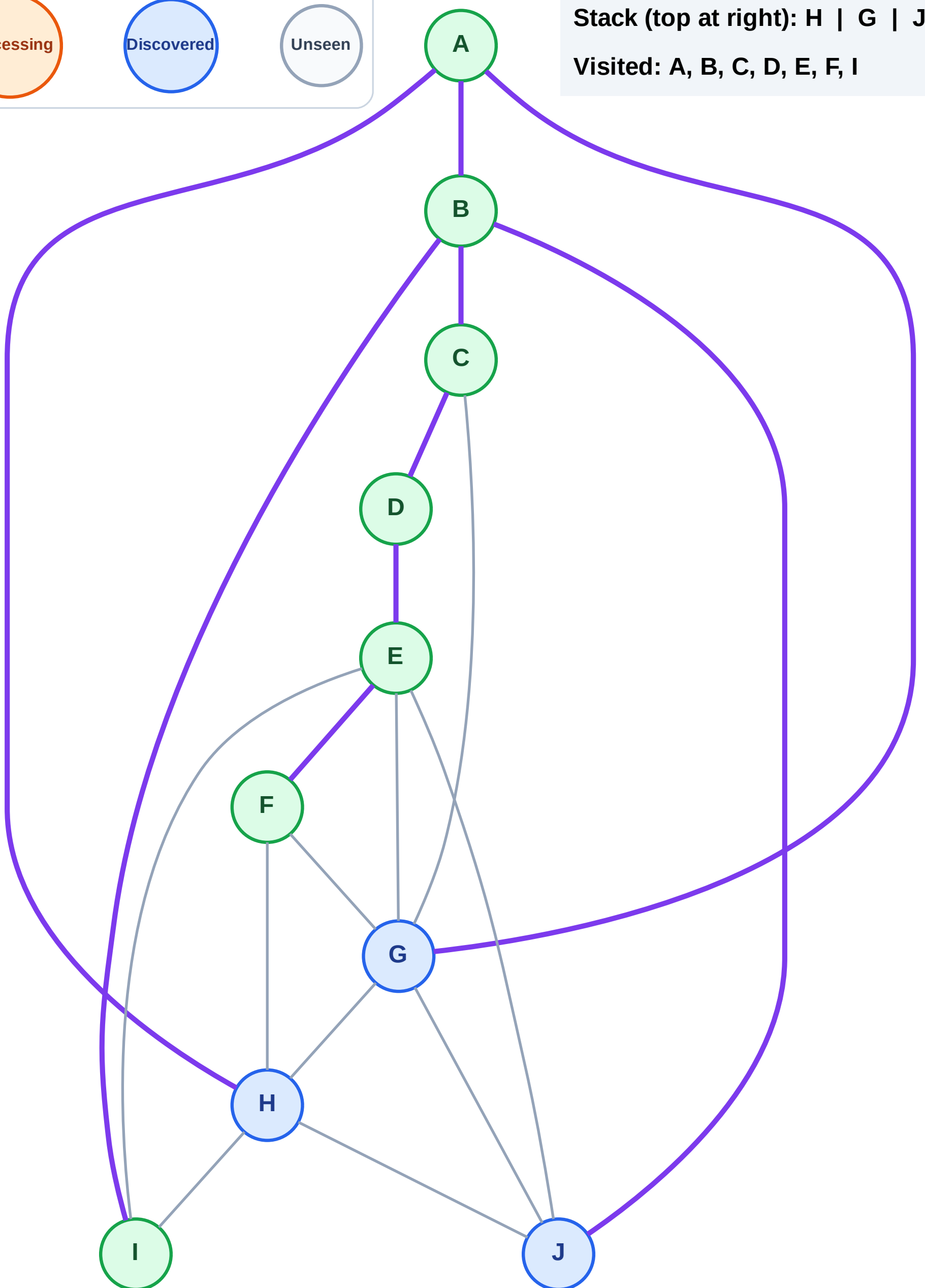
Step 15: No new neighbors from I

Legend



Stack (top at right): H | G | J

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node J

Legend

Complete

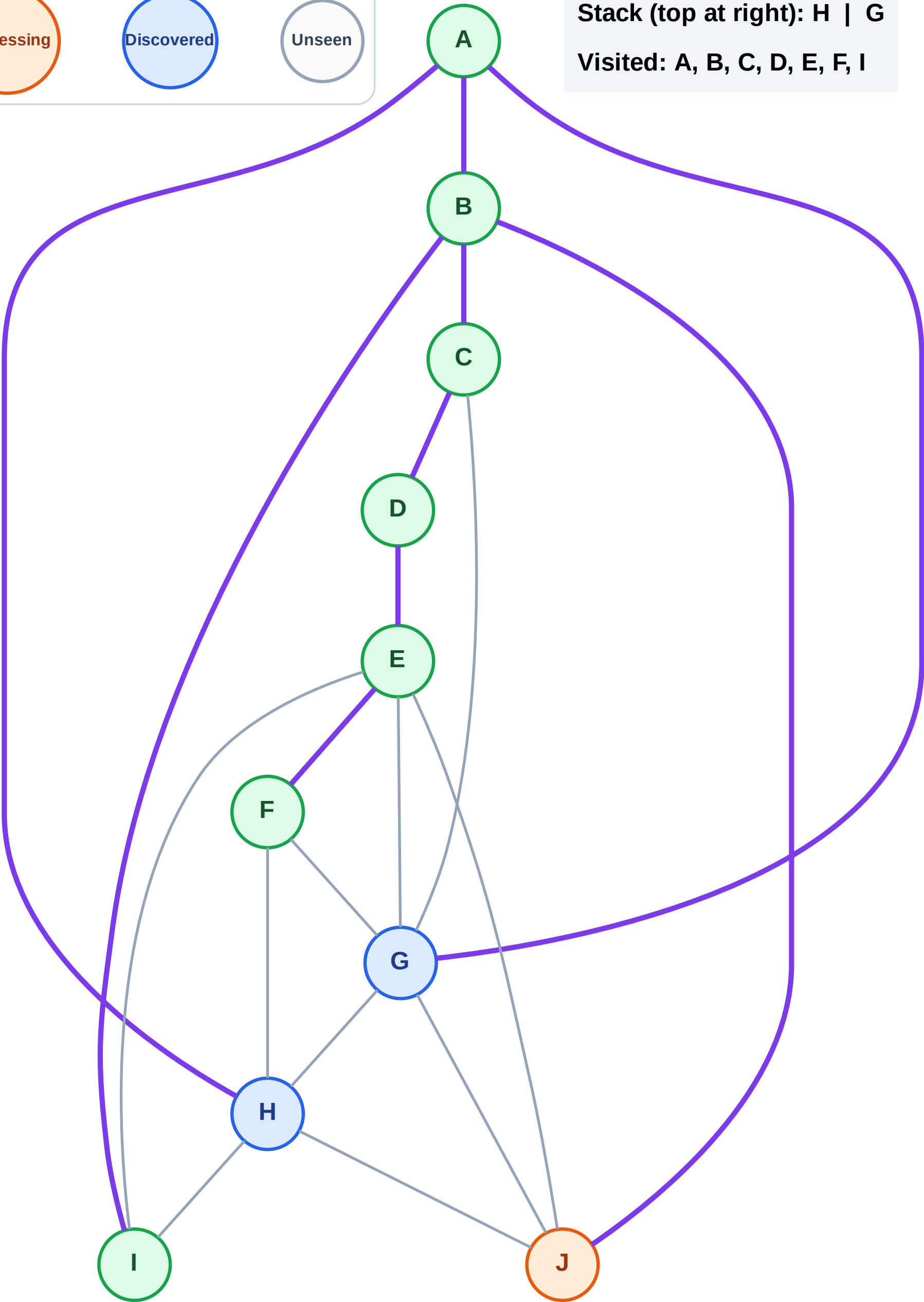
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 17: No new neighbors from J

Legend

Complete

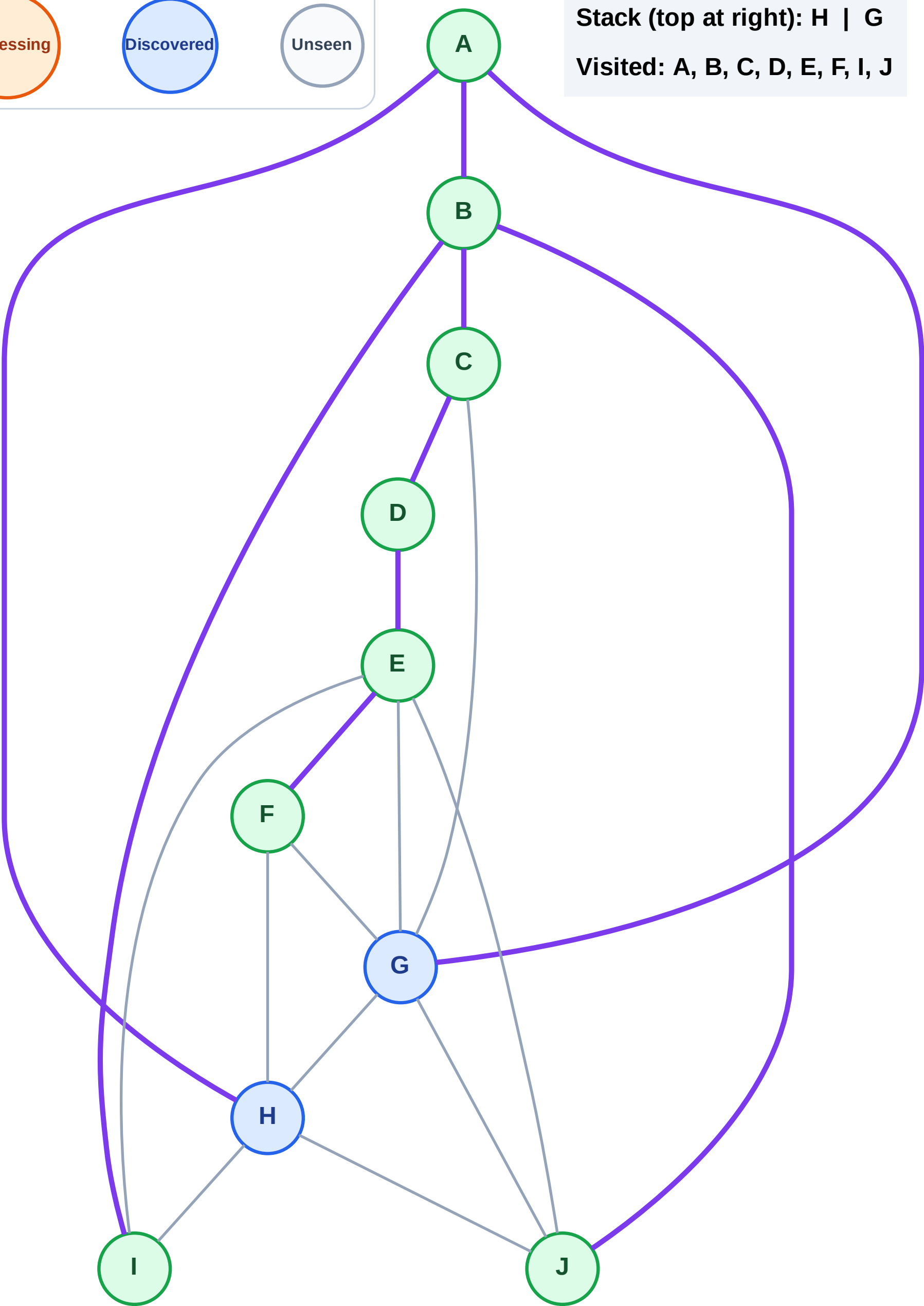
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

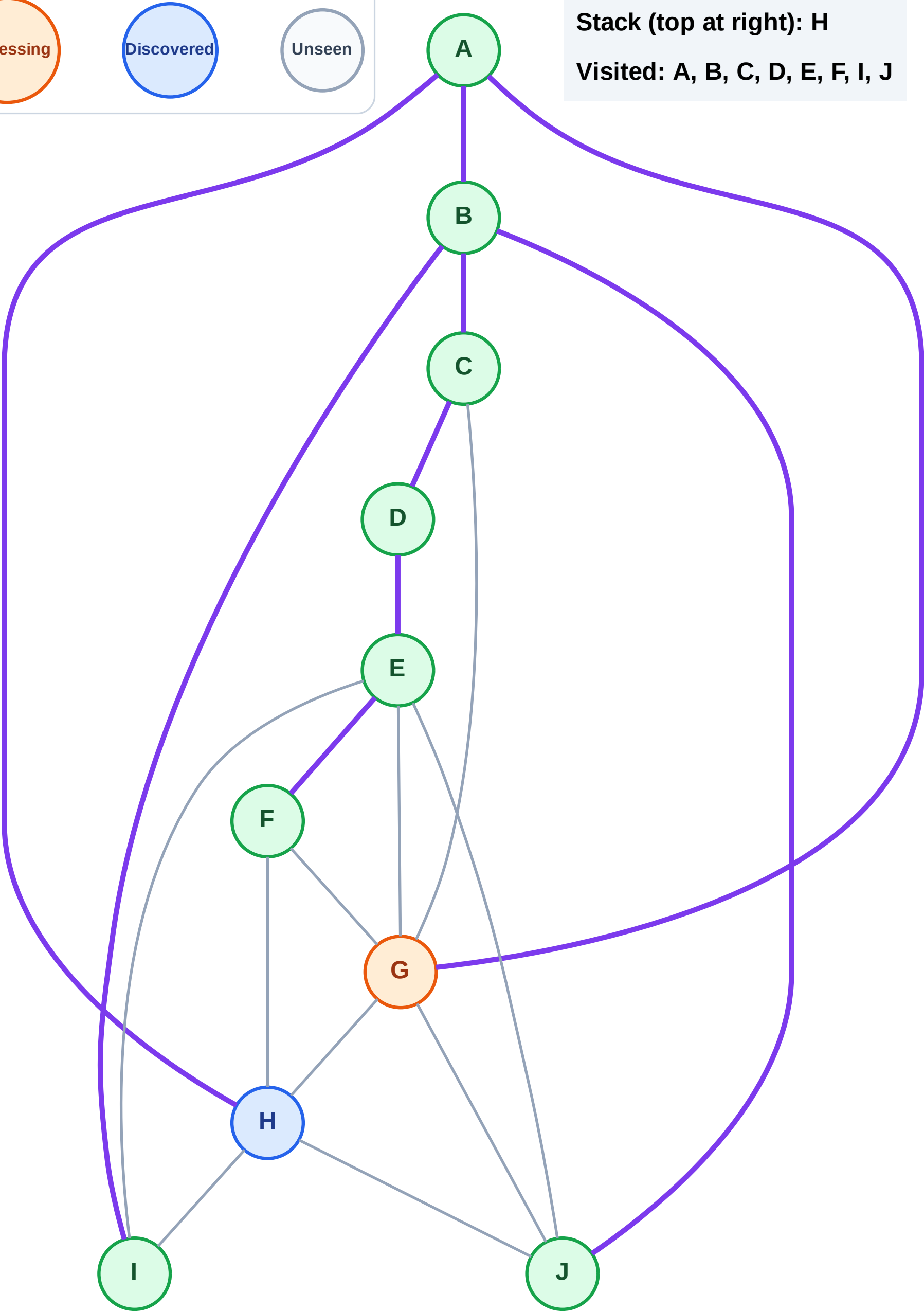
Complete

Processing

Discovered

Unseen

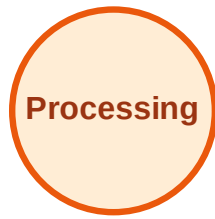
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

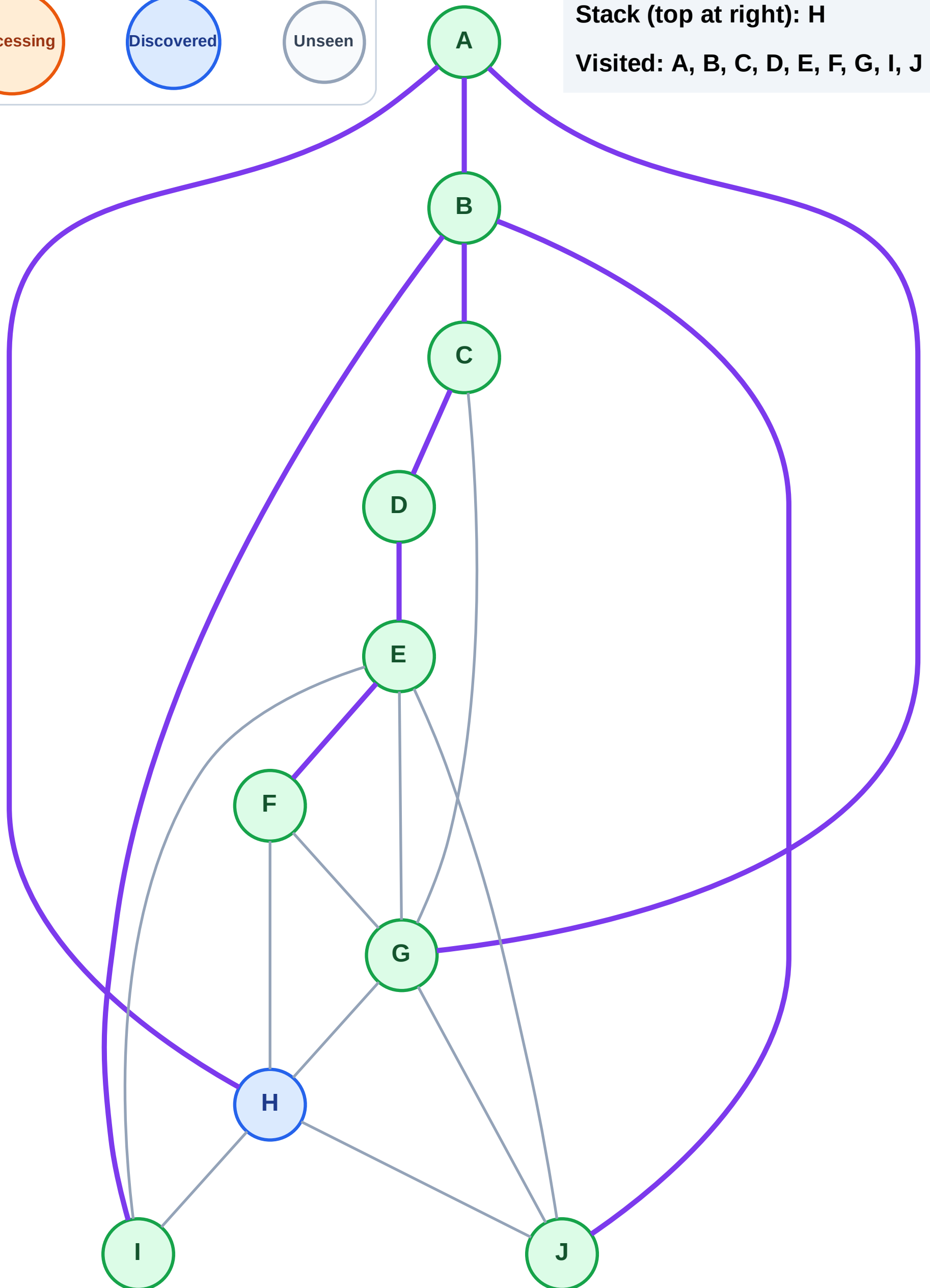
Step 19: No new neighbors from G

Legend



Stack (top at right): H

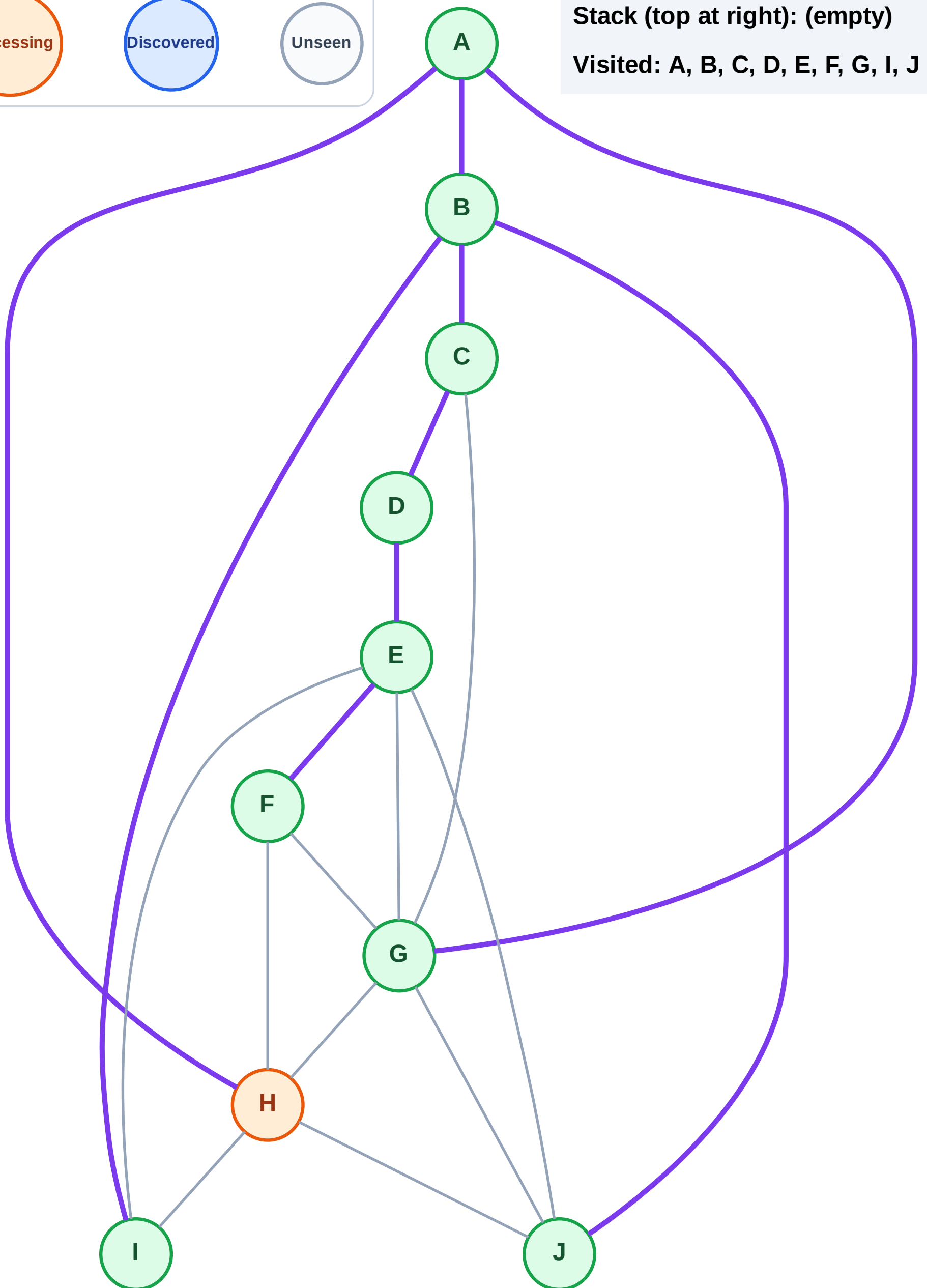
Visited: A, B, C, D, E, F, G, I, J



Step 20: Process node H



Visited: A, B, C, D, E, F, G, I, J



Step 21: No new neighbors from H

Step 21: No new neighbors from H

